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Master Thesis

Randomized Identification Codes with a Focus on K-Identification

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Topic: Randomized Identification Codes with a Focus on K-Identification

Task description:

Identification via channels (ID) asks receivers to decide whether a specific message was sent, rather than to reconstruct it. Randomized ID codes achieve doubly-exponential message sets in blocklength and enable vanishing Type-I (false-accept) and Type-II (miss) error probabilities at positive ID rate. K-Identification (K-ID) generalizes this: the receiver asks whether the transmitted message belongs to a set of K candidates. K-ID is relevant for goal/semantics-driven communication (e.g., digital twins), where the action depends only on whether the current state falls into a small set of actionable hypotheses, not on exact message recovery. This thesis will study randomized ID codes with emphasis on K-ID, and extend the ideas and evaluation from “Stopping the Data Flood: Post-Shannon traffic reduction in digital-twins applications” to the K-ID setting. The work should balance theory, algorithms, and software-driven experiments using our provided library.

The student is expected to:

Survey the state of the art:

- Core ID concepts: randomized encoders/acceptance regions, ID capacity, error exponents, and operational interpretations; contrast with Shannon transmission.
- Define K-ID precisely (membership in a K-set) and position it vs. related notions (multiple-object ID, generalized ID; deterministic vs randomized variants).

Formulate the problem & metrics for K-ID:

- Formal model (channel class, randomization/common randomness budget, decoder tests).
- Report both Type-I/Type-II errors and computational metrics (encode/decode time, seed length, memory), plus finite-blocklength behavior.

Replicate & extend the digital-twins baseline to K-ID:

- Reproduce key experiments/results from the “Stopping the Data Flood” study (baseline ID) with our software stack.
- Design K-ID experiments that reflect digital-twin quantifying traffic reduction vs. reliability/latency.

Use our software library to build a modular K-ID test bench:

- Implement randomized ID/K-ID encoders and decoders (acceptance region tests) as pluggable components.

Design and run a simulation campaign:

- Measure KPIs such as: Type-I/II error curves, empirical exponents, queries per symbol, and compute/memory usage. Quantify traffic reduction vs. reliability/latency

Deliver a comparative assessment with clear takeaways:

- When do systems benefit from K-ID?
- Tradeoffs among randomization, complexity, and reliability.

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Statement of authorship

I hereby certify that I have authored this document entitled *Randomized Identification Codes with a Focus on K-Identification* independently and without undue assistance from third parties. No other than the resources and references indicated in this document have been used. I have marked both literal and accordingly adopted quotations as such. During the preparation of this document I was only supported by the following persons:

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Additional persons were not involved in the intellectual preparation of the present document. I am aware that violations of this declaration may lead to subsequent withdrawal of the academic degree.

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Abstract

In time-critical **NCSs**, the utility of **DTs** relies strictly on their synchronization accuracy with physical assets. Traditional periodic full-state transmission, while robust, overlooks the predictive capabilities of the **DT**, resulting in significant bandwidth redundancy. To address this inefficiency, this thesis investigates **ID-based synchronization**, with a specific focus on the **K-ID** protocol for semantic compliance. Unlike classical communication aimed at data reconstruction, **K-ID** transforms the synchronization task into a semantic verification problem: confirming whether the remote state lies within a “safety tolerance set” predicted locally.

The main contributions of this thesis are threefold. First, we establish an end-to-end latency model to compare algebraic coding (**RS-ID**) and cryptographic hashing (**SHA-256**), revealing key engineering trade-offs in edge computing. Experiments demonstrate that truncated **SHA-256** offers stable encoding latency ($\approx 2.2 \mu s$) independent of payload size, making it a superior engineering choice for **URLLC** applications compared to **RS-ID**, which suffers from field saturation at short blocklengths. Second, through extensive parameter sweeps, we map the Risk-Speedup Pareto Frontier of **K-ID**, proving that introducing moderate semantic tolerance (e.g., $R = 10$) can reduce fallback frequency by an order of magnitude, significantly enhancing communication efficiency under bounded risk. Finally, we evaluate error evolution under a dynamic random walk model, uncovering the phenomenon of “Cascading Failures” missed by static models: weak verifiers ($n_{ver} = 8$) can cause single hash collisions to evolve into persistent boundary breaches. Furthermore, the heavy-tail distribution of Inter-Fallback Times (**IFT**) confirms **K-ID**’s ability to transform continuous sensor streams into sporadic bursty traffic, providing a theoretical foundation for resource scheduling in next-generation semantic communication networks.

Keywords: Digital Twins, K-Identification, Semantic Communication, URLLC, Networked Control Systems, Hash Collision, Random Walk

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Acronyms

Aol	Age of Information	10
BIBO	Bounded-Input Bounded-Output	11
CDF	Cumulative Distribution Function	39, 40, V
CPS	Cyber-Physical Systems	1
CR	Common Randomness	7
CSI	Channel State Information	42
DRBG	Deterministic Random Bit Generator	8
DT	Digital Twin	1, 3–5, 7–11, 17, 19, 20, 32, 38, 42, V, D
ETC	Event-Triggered Control	4, 5, 11
HRC	Human-Robot Collaboration	10
ID	Identification	1–3, 5, 7, 10, 21, 22, 24, 30, 32–34, 41, IV, V, D
IFT	Inter-Fallback Time	39, 40, V, D
IoT	Internet of Things	42
K-ID	K-Identification	1–3, 9–11, 17, 20, 21, 24, 30, 35, 37, 41, 42, V, D
LiDAR	Light Detection and Ranging	16, 33, 34, 41, II
NCS	Networked Control System	11, D
PLKG	Physical Layer Key Generation	8, 42
PRNG	Pseudo-Random Number Generator	8, 42
PTP	Precision Time Protocol	8, 42
ROM	Random Oracle Model	15, 16
RS-ID	Reed-Solomon Identification	2, 6–9, 16, 24, 30, 31, 33, 34, 41, VI, D
SHA-256	Secure Hash Algorithm 256-bit	2, 7, 9, 16, 24, 30, 31, 33, 34, 41, VI, D
TCP	Transmission Control Protocol	4
TDD	Time-Division Duplex	42
UDP	User Datagram Protocol	4
URLLC	Ultra-Reliable Low-Latency Communications	1, 3, 4, 41, D
Vol	Value of Information	10
ZOH	Zero-Order Hold	20, 43

1 Introduction

1.1 Background and Motivation

DTs have become a standard component in modern industrial automation for connecting physical assets with their virtual counterparts. By maintaining accurate virtual copies of systems like industrial robots, **DTs** enable real-time monitoring, predictive maintenance, and remote control. In time-critical **Cyber-Physical Systems (CPS)** applications, the utility of a **DT** is strictly defined by the synchronization accuracy between the physical robot and the digital model [22]. Consequently, ensuring state consistency typically demands **Ultra-Reliable Low-Latency Communications (URLLC)** capabilities to support high-frequency updates.

However, meeting these requirements via standard periodic transmission is both resource-intensive and inefficient. Since the **DT** often possesses a precise estimate of the robot’s state using its internal kinematic model, continuously transmitting raw telemetry data creates significant redundancy. This “blind” transmission consumes scarce bandwidth resources without providing meaningful information gain.

To mitigate this redundancy, recent research proposes identification-based **ID** synchronization [14]. Unlike classical Shannon-type communication, which aims to reconstruct the full message, **ID** schemes focus on a verification task: confirming whether the remote state matches a local hypothesis. Building on this concept, prior works [18] demonstrate that transmitting a compressed “verifier” (e.g., a short hash or randomized code) instead of the full state can significantly reduce communication traffic. This aligns with the paradigm of semantic communication in 6G networks [7], shifting the focus from what data is sent to what value the data holds for control.

Despite its theoretical appeal, deploying **ID** synchronization in real-time control loops reveals two critical engineering trade-offs that serve as the primary motivation for this thesis. First, regarding the latency trade-off, the computational overhead of encoding on edge devices may exceed the time saved by reducing transmission traffic. This shift from a communication-bound to a computation-bound bottleneck necessitates a rigorous end-to-end latency analysis. Second, reliability becomes a primary concern when moving from exact to semantic verification. In practical control, enforcing bitwise synchronization (standard **ID**) is often overly conservative, triggering unnecessary updates for benign deviations such as sensor noise. We need to extend verification to semantic “safety sets”, i.e., **K-ID**. However, this relaxation introduces collision risks, specifically Type II (miss) errors, where dangerous desynchronizations are incorrectly classified as safe.

1.2 Problem Statement

Existing research on [ID](#) has primarily focused on information-theoretic capacity or simple traffic suppression, lacking a comprehensive evaluation of latency, physical safety, and computational feasibility. Specifically, it remains unclear under what bandwidth and computational constraints [ID](#) truly outperforms traditional transmission, and how to quantify the Pareto trade-off between semantic tolerance and safety risk in [K-ID](#) systems.

1.3 Contributions

To address these limitations, this thesis proposes a comprehensive framework ranging from theoretical modeling to dynamic evaluation, investigating not only the static identification problem but also error evolution in dynamic environments.

First, we develop an end-to-end latency model combining computational and transmission delays to experimentally compare algebraic coding ([RS-ID](#)) and cryptographic hashing ([SHA-256](#)). The results establish that truncated [SHA-256](#) is the superior engineering choice for edge deployment, offering stable encoding latency ($\approx 2.2 \mu s$) independent of payload size, whereas [RS-ID](#) suffers from severe field saturation at short blocklengths.

Second, we generalize the identification problem to [K-ID](#), introducing the concept of Semantic Tolerance to allow the system to filter benign jitter. Through extensive parameter sweeps, we map the Risk-Speedup Pareto Frontier, demonstrating that introducing moderate tolerance can reduce fallback frequency by an order of magnitude while maintaining bounded safety risks. This effectively trades data freshness for operational stability.

Third, we extend the evaluation to a time-correlated Random Walk model. The experiments reveal the phenomenon of “Cascading Failures” which static models fail to capture: weak verifiers ($n_{ver} = 8$) can cause single hash collisions to evolve into persistent, undetected boundary breaches. Furthermore, the analysis of Inter-Fallback Times confirms the ability of [K-ID](#) to transform continuous sensor streams into sporadic bursty traffic, providing a theoretical basis for network scheduling.

1.4 Thesis Organization

The remainder of this thesis is organized as follows. Chapter [2](#) reviews related work, covering the transition from data reconstruction to semantic identification, the mathematical principles of encoding mechanisms, and the control-theoretic background. Chapter [3](#) details the system model, data flow design, and the derivation of theoretical error bounds for [K-ID](#). Chapter [4](#) describes the implementation of the modular Python simulation framework and the experimental design strategy. Chapter [5](#) presents the experimental results, including latency sensitivity analysis, Pareto optimization of [K-ID](#), and risk assessment under dynamic drift scenarios. Finally, Chapter [6](#) concludes the thesis and outlines future research directions.

2 Background and Related Work

This chapter reviews the transition from classical data transmission to identification-based synchronization, positioning it within the broader context of semantic communication [23] for 6G networks. We analyze alternative synchronization strategies—such as differential encoding and event-triggered control—to motivate the engineering need for the identification (ID) paradigm. We then compare practical encoding mechanisms by contrasting the avalanche dynamics of cryptographic hashing with the geometric properties of algebraic coding. Finally, we discuss how this paradigm generalizes to K-ID, where strict equality constraints are relaxed to semantic safety sets to accommodate natural variability in physical control loops.

2.1 From Data Reconstruction to Semantic Identification

In time-critical control loops, the utility of a DT depends on its synchronization with the physical asset. Conventional protocols treat this as a data reconstruction problem, where the robot periodically transmits its full state vector S_t (e.g., joint angles and velocities) to the DT. While robust, this approach overlooks a key redundancy: the DT, using its internal kinematic model, already maintains a prediction $\hat{S}_t$ of the robot’s state. The communication task is therefore not to inform the DT “where the robot is,” but to verify “whether the DT’s prediction is correct.”

This shift aligns with the paradigm of semantic communication [20] in 6G networks. Unlike classical Shannon-type communication, which emphasizes accurate reconstruction, semantic communication emphasizes information that matters for the receiver’s goal. In the context of URLLC, this suggests prioritizing “actionable” information—such as safety violations—over raw telemetry.

Accordingly, the coding objective shifts from Shannon’s transmission capacity to the identification capacity defined by Ahlswede and Dueck [3]. In transmission, each possible state must map to a disjoint decoding set, limiting the number of distinguishable states to 2^{nR} for blocklength n . In contrast, identification permits overlaps between decoding sets, as long as pairwise intersections are statistically negligible. This relaxation enables a doubly exponential growth in identifiable states ($2^{2^{nR}}$), meaning that short verifiers can discriminate among an extremely large state space while allowing a vanishingly small error probability.

2.2 Alternative Synchronization Strategies

Before detailing the implementation of identification, it is necessary to rigorously evaluate why established data reduction strategies—specifically Delta Encoding and Event-Triggered Control—are often insufficient for unreliable edge networks.

2.2.1 Full State Transmission (The Baseline)

The standard baseline in industrial automation is the periodic transmission of the full state vector S_t . This approach is favored for its statelessness; each packet is self-contained. If a packet at time t is lost due to channel interference, the packet at $t + 1$ allows the DT to immediately recover the correct state. However, this robustness comes at the cost of extreme bandwidth inefficiency. In steady-state operations, where $S_t \approx S_{t-1}$, the information entropy of the transmitted packet is near zero, yet it consumes the full bandwidth allocated to the payload n_{data} .

2.2.2 Delta Encoding and the Drift Problem

To address this redundancy, Delta Encoding [16] (or differential transmission) is frequently employed. This method transmits only the difference vector $\Delta_t = S_t - S_{t-1}$. Since Δ_t typically has a smaller dynamic range than the absolute state S_t , it can be quantized with fewer bits, offering significant compression ratios.

However, Delta Encoding introduces a key vulnerability: statefulness. The reconstruction of the current state at the receiver, denoted as S'_t , depends recursively on the history of received updates: $S'_t = S'_{t-1} + \Delta_t$. This recursive dependency creates a clear failure mode in unreliable transport protocols like User Datagram Protocol (UDP). A single lost packet Δ_k causes a persistent offset between the physical robot and the DT, a phenomenon known as “state drift.” [12] While reliable protocols like Transmission Control Protocol (TCP) can prevent packet loss, retransmissions introduce non-deterministic latency jitter, which can violate the timing deadlines of real-time control loops. Therefore, Delta Encoding poses a high risk for URLLC applications.

2.2.3 Event-Triggered Control (ETC)

Another prominent alternative is Event-Triggered Control (Event-Triggered Control (ETC)). Unlike Time-Triggered (periodic) systems, ETC only transmits data when the deviation between the current state and the last transmitted state exceeds a threshold δ : $\|S_t - S_{last_sent}\| > \delta$ [21]. This approach effectively silences the network during periods of inactivity.

Nevertheless, ETC relies fundamentally on a “push” communication model. A silent channel is ambiguous to the receiver: it could indicate that the system is operating safely within bounds, or it could indicate that the sensor has failed or the link has been severed. In safety-critical robotics, such ambiguity is often unacceptable. Furthermore, ETC requires the sensing node to perform continuous monitoring and threshold checking, pushing computational complexity to the energy-constrained edge [8].

2.2.4 The Identification Advantage

In contrast to these alternatives, the **ID** scheme proposed in this thesis combines statelessness with strong compression. Unlike Delta Encoding, **ID** verification checks the absolute state S_t directly and is therefore immune to drift; if a packet is lost, the next packet again provides an independent check. Unlike **ETC**, **ID** supports a “verify” model, allowing the **DT** to actively confirm synchronization with minimal overhead. This combination is well-suited for maintaining high-fidelity **DTs** over unreliable networks.

2.3 Encoding Mechanisms: Mathematical Structure and Bias

To realize the theoretical capacity of identification in edge computing, the encoding process must map a high-dimensional state space $\mathcal{S}$ into a low-dimensional verifier space $\mathcal{V}$ while minimizing collision probabilities. This section dissects the internal mechanisms of the two candidate algorithms—cryptographic hashing (SHA-256) and algebraic coding (RS-ID)—to reveal the mathematical origins of their reliability.

2.3.1 Cryptographic Construction: The SHA-256 Architecture

To understand the source of the high entropy in the generated verifiers, it is necessary to examine the internal structure of the SHA-256 algorithm [17]. SHA-256 belongs to the SHA-2 family and is built upon the Merkle-Damgård construction, which iteratively processes the input state S using a dedicated compression function. The process consists of three mathematically distinct stages: preprocessing, message expansion, and the state update loop.

Preprocessing and Padding

The input message M (comprising the state S concatenated with the seed r) is first padded to ensure its length is a multiple of 512 bits. The padding scheme appends a single ‘1’ bit, followed by k zero bits, and finally a 64-bit integer representing the original message length. This padded message is parsed into N blocks of 512 bits, denoted as $M^{(1)}, M^{(2)}, \dots, M^{(N)}$.

Message Schedule (Expansion)

The core of the diffusion process begins with the Message Schedule. Each 512-bit block $M^{(i)}$ is expanded from 16 words (32-bit each) into 64 words, denoted as $W_0, \dots, W_{63}$. The first 16 words are copied directly from the message block. The remaining 48 words are generated recursively, creating a complex dependency chain where every future bit depends on multiple past bits:

$$W_t = \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16} \pmod{2^{32}} \quad (2.1)$$

where the small sigma functions σ_0 and σ_1 introduce the initial bitwise diffusion via rotation (ROTR) and shifting (SHR):

$$\sigma_0(x) = \text{ROTR}^7(x) \oplus \text{ROTR}^{18}(x) \oplus \text{SHR}^3(x) \quad (2.2)$$

$$\sigma_1(x) = \text{ROTR}^{17}(x) \oplus \text{ROTR}^{19}(x) \oplus \text{SHR}^{10}(x) \quad (2.3)$$

This expansion ensures that a single bit flip in the input $M^{(i)}$ propagates to affect a vast majority of the 64 words in the schedule, setting the stage for the avalanche effect.

The Compression Loop

The 64 expanded words W_t are then fed into the compression function. The working state consists of eight 32-bit variables, labeled a through h , initialized to constant prime-root values. For each step t from 0 to 63, the state is updated using two temporary variables T_1 and T_2 :

$$T_1 = h + \Sigma_1(e) + Ch(e, f, g) + K_t + W_t \quad (2.4)$$

$$T_2 = \Sigma_0(a) + Maj(a, b, c) \quad (2.5)$$

Here, K_t represents 64 constant 32-bit words. The non-linear logical functions Ch (Choose) and Maj (Majority), along with the big sigma functions Σ_0 and Σ_1 , serve as the mixing engine:

$$Ch(x, y, z) = (x \wedge y) \oplus (\neg x \wedge z) \quad (2.6)$$

$$Maj(x, y, z) = (x \wedge y) \oplus (x \wedge z) \oplus (y \wedge z) \quad (2.7)$$

$$\Sigma_0(x) = ROTR^2(x) \oplus ROTR^{13}(x) \oplus ROTR^{22}(x) \quad (2.8)$$

$$\Sigma_1(x) = ROTR^6(x) \oplus ROTR^{11}(x) \oplus ROTR^{25}(x) \quad (2.9)$$

After calculating T_1 and T_2 , the registers are shifted ($h \leftarrow g, g \leftarrow f, \dots$), and updated as:

$$e \leftarrow d + T_1, \quad a \leftarrow T_1 + T_2 \quad (2.10)$$

This rigorous mixing guarantees that after 64 rounds, every bit of the final hash output is a complex, non-linear function of every bit of the input block. Finally, the output verifier v is obtained by truncating this digest to the target length n_{ver} , retaining the high-entropy properties generated by this procedure.

2.3.2 Algebraic Construction: The Geometry of RS-ID

In contrast to the bitwise diffusion of SHA-256, [RS-ID](#) relies on the geometric properties of polynomials over finite fields[19].

Data as a Polynomial Curve

Let the state vector S be decomposed into k segments $\{s_0, s_1, \dots, s_{k-1}\}$, where each segment is an element of the Galois Field $\mathbb{F}_q$. We interpret these segments not as raw data, but as coefficients of a unique polynomial curve $P_S(x)$ of degree $k - 1$:

$$P_S(x) = s_0 + s_1x + s_2x^2 + \dots + s_{k-1}x^{k-1} = \sum_{i=0}^{k-1} s_i x^i \quad (2.11)$$

The generation of a verifier v using a random seed r is geometrically equivalent to sampling this curve at the coordinate $x = r$:

$$v = P_S(r) \pmod{q} \quad (2.12)$$

Collision as Intersection

A collision (Type II error) occurs when the true state S and the predicted state $\hat{S}$ are distinct ($S \neq \hat{S}$), yet their verifiers match ($v = \hat{v}$). In our geometric model, this implies:

$$P_S(r) = P_{\hat{S}}(r) \implies P_S(r) - P_{\hat{S}}(r) = 0 \quad (2.13)$$

Let $\Delta(x) = P_S(x) - P_{\hat{S}}(x)$ be the difference polynomial. Since $S \neq \hat{S}$, $\Delta(x)$ is a non-zero polynomial with degree at most $k - 1$. By the Fundamental Theorem of Algebra, a non-zero polynomial of degree d can have at most d roots.

This geometric constraint provides the rigorous upper bound for the collision probability. The number of "bad seeds" r that cause a collision is limited to the number of intersection points between the two curves $P_S(x)$ and $P_{\hat{S}}(x)$, which is at most $k - 1$. Thus, for a field of size q , the probability is:

$$P_{miss} = \frac{\text{Number of Roots}}{q} \leq \frac{k - 1}{q} \quad (2.14)$$

This derivation visualizes the identification process as "checking for intersection." Since two distinct curves cannot intersect everywhere, checking a random point r effectively distinguishes them with high probability [19].

2.3.3 Structural Blindness and Randomization

A critical limitation of deterministic compression is "structural blindness." Consider a simplified case where the encoder uses a modulo operation. If the error $e = S - \hat{S}$ happens to be a multiple of the modulus, the generated verifiers will be identical ($v = \hat{v}$) regardless of the magnitude of the error. Even if the robot's state S changes continuously, this error pattern might persist, leading to a persistent Type II (miss) error.

To eliminate structural blindness, identification theory mandates the use of common randomness—a shared seed r that changes with each transmission. In [RS-ID](#), changing r corresponds to sampling the polynomial at a different x -coordinate. In [SHA-256](#), appending a time-varying salt ($S||r$) changes the avalanche dynamics. By varying r , the mapping changes at every time step; although a collision may occur at a specific seed r_1 , the probability that the same error causes a collision at a new seed r_2 is independent. This converts potential "permanent failures" into "transient probabilistic events."

2.4 Operational Prerequisite: Common Randomness Generation

As established in Section 2.3, the theoretical validity of identification codes hinges on the availability of [Common Randomness \(CR\)](#)—a shared source of entropy between the encoder (robot) and the decoder ([DT](#)). Without a dynamic shared seed r , the system reduces to a deterministic mapping and becomes susceptible to structural blindness, where persistent error patterns can remain undetectable [14]. Therefore, the practical deployment of [ID](#) requires a robust mechanism to generate or synchronize r without negating the bandwidth savings.

The most straightforward approach to establish this shared entropy is to use synchronized **Pseudo-Random Number Generators (PRNGs)**, formally standardized as **Deterministic Random Bit Generators (DRBGs)** [5]. By initializing identical generators with a pre-shared secret key and tying them to a synchronized time index—typically maintained via the **Precision Time Protocol (PTP)** [11]—both the robot and the **DT** can locally derive an identical sequence of seeds r_t for each time slot. While this method has zero communication overhead during steady-state operation, it introduces a dependency on clock synchronization. Any drift in the robot’s clock relative to the **DT** results in a seed mismatch ($r_{robot} \neq r_{DT}$), causing verifier checks to fail continuously. This loss of synchronization can manifest as a sustained burst of false positives, forcing the system into a fallback mode until resynchronization.

Alternative approaches, such as **Physical Layer Key Generation (PLKG)**, attempt to extract randomness from channel reciprocity in wireless networks [15]. While **PLKG** aligns with physical-layer security paradigms by exploiting channel noise entropy [2], it often requires additional reconciliation overhead to correct quantization mismatches. Consequently, to isolate the performance characteristics of the encoding mechanisms, this thesis assumes an idealized PRNG-based synchronization model for the experimental evaluation.

2.5 Theoretical Limits and Algorithmic Trade-offs

While the introduction of randomization allows for drastic compression, the output length cannot be reduced arbitrarily. The choice between algebraic (RS-ID) and cryptographic (SHA-256) implementations is governed by a fundamental tension between theoretical error bounds and computational efficiency.

2.5.1 The Algebraic Limit: Saturation and Field Size

In the polynomial model of **RS-ID**, the verifier length L is constrained by the algebraic properties of the state representation. Since a state vector of length k is interpreted as a polynomial of degree $k - 1$, two distinct state polynomials can intersect at most at $k - 1$ points. To ensure reliable identification, the size of the evaluation field $Q = 2^L$ must be significantly larger than the number of potential intersection points. If $Q < k$, the “collision points” can saturate the field, making it mathematically impossible to find a seed r that distinguishes the two states via a single sample.

Therefore, a necessary condition for the reliability of algebraic identification is that the field size must grow with the data size:

$$2^L > k \quad \Rightarrow \quad L > \log_2 k \quad (2.15)$$

This inequality represents the physical manifestation of the double exponential limit established by Ahlswede and Dueck [3]. It dictates that for **RS-ID**, the verifier length L must scale logarithmically with the data size k . As long as this condition is met, **RS-ID** provides a rigorous upper bound λ_2 on the collision probability, ensuring that no pair of states exceeds a worst-case error threshold regardless of the input distribution [14].

2.5.2 The Cryptographic Perspective: Independence from Data Size

In contrast to the algebraic constraints, cryptographic methods like [SHA-256](#) operate under the Random Oracle Model. The collision probability of a truncated hash is approximately 2^{-L} , which is invariant to the input data size k . Unlike [RS-ID](#), [SHA-256](#) does not suffer from “saturation” when the input data grows. A 16-bit hash verifier has the same uniform collision probability whether verifying a 100-byte state or a 1-megabyte state. This makes cryptographic approaches robust for large-scale data vectors where satisfying $2^L > k$ might require an impractically large field for [RS-ID](#).

2.5.3 Engineering Trade-offs: Bound vs. Speed

From a safety-critical perspective, the algebraic structure of [RS-ID](#) provides a deterministic guarantee. Based on the Fundamental Theorem of Algebra, the number of collision-causing seeds for a message of length k is bounded by the polynomial degree $k - 1$ [14]. This yields the following worst-case bound on the miss probability:

$$P_{miss} \leq \frac{k - 1}{q} \quad (2.16)$$

where q is the field size. Such a worst-case guarantee is relevant for mission-critical control loops in which adversarial or pathological error distributions must be handled analytically rather than statistically.

The same bound also highlights a scalability limitation: the number of potentially colliding seeds grows linearly with data size ($P_{miss} \propto k$). For large payloads (e.g., LiDAR or vision streams), preserving a small upper bound requires increasing q , which in turn increases the latency of finite-field arithmetic. In contrast, cryptographic hashing (e.g., [SHA-256](#)) is commonly modeled as a random oracle, where a truncated verifier of length L yields a collision probability on the order of 2^{-L} that does not depend on k . This makes hash-based approaches attractive when the payload size dominates system constraints.

Accordingly, this thesis evaluates both mechanisms to cover the spectrum of edge computing requirements: [RS-ID](#) is used as a theoretical reference for lightweight control signals (small n_{data}), while [SHA-256](#) is used to demonstrate computational scalability for data-intensive applications (large n_{data}).

2.6 Semantic Verification: Adapting to Control Flexibility

While standard identification provides a rigorous method for exact state verification, its binary nature—accepting only a single, precise value—often contradicts the operational flexibility required in modern robotic control. In many practical scenarios, the [DT](#) does not need to enforce a strict equality constraint ($x = \hat{x}$) but rather needs to guarantee that the robot operates within a permissible semantic range. This nuance necessitates the adoption of [K-ID](#), which generalizes the verification target from a unique point to a semantic safety set.

2.6.1 The Autonomous Driving Scenario

Consider a practical example of an autonomous robot (or vehicle) navigating a public roadway. The control policy might specify a safe velocity corridor, such as “maintain speed between 40 km/h and 60 km/h.” If the robot moves at 45 km/h while the **DT** predicted 50 km/h, a standard **ID** scheme would flag this as a collision (mismatch) because the hashed values of “45” and “50” are distinct. This triggers an unnecessary data transmission to correct a behavior that is semantically valid.

K-ID resolves this inefficiency by defining the acceptance region $\mathcal{B}$ as the interval $[40, 60]$. As long as the robot’s state falls within this set, the **K-ID** verifier confirms validity without consuming bandwidth for minor deviations.

2.6.2 Human-Robot Collaboration and Jitter

This logic extends beyond simple velocity constraints to more complex spatial and interactive scenarios. For instance, in **Human-Robot Collaboration (HRC)**, the exact coordinates of a robotic arm are less critical than ensuring the end-effector remains outside a defined “human safety bubble” [22]. Similarly, for high-precision assembly tasks involving compliant mechanisms, the robot may exhibit minor high-frequency vibrations (jitter) that are mechanically acceptable. Standard **ID** would treat these micro-vibrations as errors, whereas **K-ID** can be tuned to ignore high-frequency noise while still detecting significant trajectory deviations.

By shifting the verification paradigm from data exactness to semantic compliance, **K-ID** aligns the synchronization protocol with the actual control objectives. It allows the system to filter out benign variability—whether it be speed fluctuations, sensor noise, or compliant jitter—and focus limited communication resources on detecting genuine safety violations. This thesis specifically investigates the implementation of such semantic verification using randomized encoding on edge constraints.

2.6.3 Discussion: Trading Freshness for Value

The scenarios described above illustrate a deviation from traditional network optimization. Conventional real-time protocols often aim to minimize the **Age of Information (AoI)** [13], keeping the **DT**’s estimate as fresh as possible regardless of content. Under an **AoI**-minimization strategy, the robot would transmit its velocity continuously even when cruising steadily within the safe corridor, purely to prevent the timestamp from “aging.”

K-ID challenges this paradigm by introducing a semantic filter that prioritizes the **Value of Information (VoI)** over simple freshness. As demonstrated in the autonomous driving case, a “stale” position estimate (high **AoI**) that remains within the safety tolerance $\mathcal{B}$ retains its utility for the control loop. By allowing **AoI** to increase during semantically safe intervals, **K-ID** trades unnecessary freshness for bandwidth efficiency. This is consistent with goal-oriented communication [20], where the objective shifts from minimizing average latency to minimizing the latency of meaningful (unsafe) updates.

2.7 Control-Theoretic Perspectives: Deadbands and Stability

Beyond communication efficiency, the deployment of **K-ID** changes the control loop structure. From a control-theoretic perspective, **K-ID** yields a **Networked Control System (NCS)** operating under a semantic sampling policy. This section analyzes the implications for stability and execution semantics, and maps **K-ID** parameters to established control concepts.

2.7.1 K-ID as Deadband Control

The tolerance mechanism in **K-ID**, defined by the acceptance set $\mathcal{B}$, is mathematically equivalent to Deadband Control (or Dead-Zone Control) in non-linear systems theory. In classical mechanics, deadbands are often viewed as parasitic nonlinearities (e.g., gear backlash) that degrade performance. However, in resource-constrained **NCS**, artificially introduced deadbands are a recognized strategy for reducing actuator wear and communication load [4].

By enforcing a “silence” policy when the synchronization error $e_t = S_t - \hat{S}_t$ satisfies $|e_t| \leq K$, **K-ID** implements Lebesgue Sampling (sampling based on value domains), as opposed to the traditional Riemann Sampling (sampling based on time domains). This semantic filtering prevents the control loop from reacting to high-frequency, low-amplitude noise (chattering), thereby reducing mechanical stress on physical actuators while preserving bandwidth for significant trajectory corrections.

2.7.2 Prevention of Zeno Behavior

A pervasive challenge in standard Event-Triggered Control (**ETC**) is the risk of Zeno behavior—a phenomenon where the inter-event time converges to zero, causing the system to trigger an infinite number of updates in a finite time interval [8]. This can occur when the state trajectory “slides” along the triggering threshold, potentially leading to computational livelock or network storms.

K-ID inherently circumvents this instability through its hybrid architecture. Although the decision to update is event-driven (based on the semantic check $v \in \mathcal{V}_K$), the verification opportunity occurs at discrete, time-triggered intervals (ticks). This imposes a strict lower bound on the inter-transmission time, denoted as $T_{min} \geq T_{tick}$. By structural design, **K-ID** guarantees Zeno-freeness, ensuring that the network load remains deterministically bounded even in worst-case divergence scenarios.

2.7.3 Bounded-Input Bounded-Output (BIBO) Stability

The introduction of the semantic tolerance K raises the question of closed-loop stability. Under **K-ID**, the **DT** (controller) operates on an estimated state $\hat{S}_t$ that deviates from the true plant state S_t by an error term δ_t . Assuming the absence of collision-induced miss errors (which are probabilistically negligible as shown in Section 2.3), the synchronization error is strictly bounded:

$$\|\delta_t\| = \|S_t - \hat{S}_t\| \leq K \quad (2.17)$$

From a stability analysis perspective, this bounded error acts as a bounded disturbance input to the controller. According to the principle of **Bounded-Input Bounded-Output (BIBO)**

stability [10], as long as the underlying control law can reject disturbances of magnitude K , the closed-loop system remains stable. This implies that the design of K is not only a traffic parameter but also a stability constraint: K must remain within the controller's stability margin (or region of attraction).

3 Design Methodology

This chapter formulated the system model for state synchronization, advancing from the classical transmission baseline to the identification (ID) and tolerance-based K-ID paradigms. To quantify the performance gains, we established an analytical latency model comparing these approaches. Furthermore, we dissected the K-ID mechanism, analyzing the critical engineering trade-offs involving quantization resolution, computational cost (contrasting RS-ID and SHA-256), and the collision risks associated with the acceptance set size K . Finally, the analysis extended to a dynamic random walk model, providing a theoretical basis for probing safety boundaries and error accumulation in time-varying systems

3.1 System Model and Data Flow

We consider a networked control system where a Robot (Sender) synchronizes its state S_t with a Digital Twin (Receiver) maintaining a prediction $\hat{S}_t$. The fundamental difference between the two approaches lies in their data processing pipelines.

3.1.1 Baseline: Traditional Transmission Flow

The standard synchronization protocol follows a linear, “always-on” approach. As illustrated in Figure 3.1, the robot first samples its sensors to generate the full state message S_t (size n_{data}) and transmits the entire payload over the channel. The Digital Twin receives the message, overwrites its local prediction $\hat{S}_t$ with the received S_t , and confirms the update.

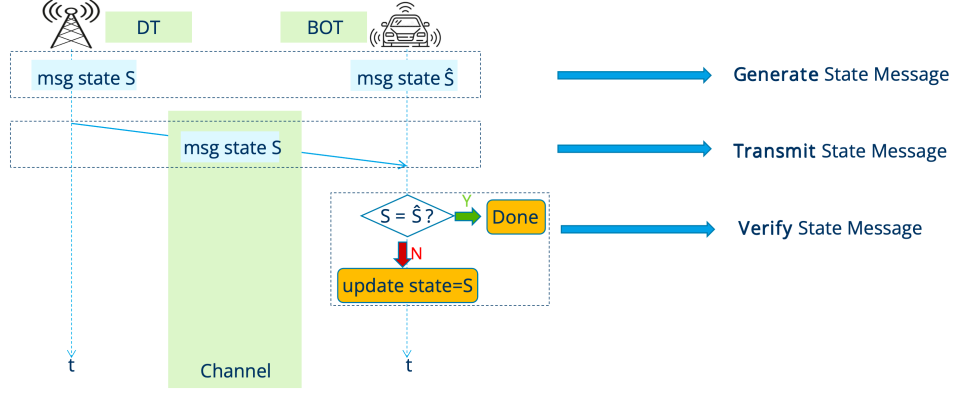


Figure 3.1: Data flow of Traditional Synchronization: The full state is transmitted regardless of the Digital Twin’s knowledge.

This process is robust but inefficient, since the transmission cost is paid unconditionally. Under this baseline, the transmission latency is deterministic and scales directly with the state size, $T_{trans} = n_{data}/B$.

3.1.2 Proposed: Identification (ID) Data Flow

The identification-based protocol reduces communication load by verifying consistency before sending the full state. As shown in Figure 3.2, both the Robot and the DT first generate their respective state views, S_t and $\hat{S}_t$. The Robot computes a compact verifier $v = E(S_t, r)$ using a shared seed r , while the DT computes the corresponding reference $\hat{v} = E(\hat{S}_t, r)$ locally. Only the short verifier v (size $n_{ver} \ll n_{data}$) is transmitted. The DT then compares v and $\hat{v}$: if they match, the protocol terminates; if they differ, the DT triggers fallback and requests the full payload S_t to resynchronize.

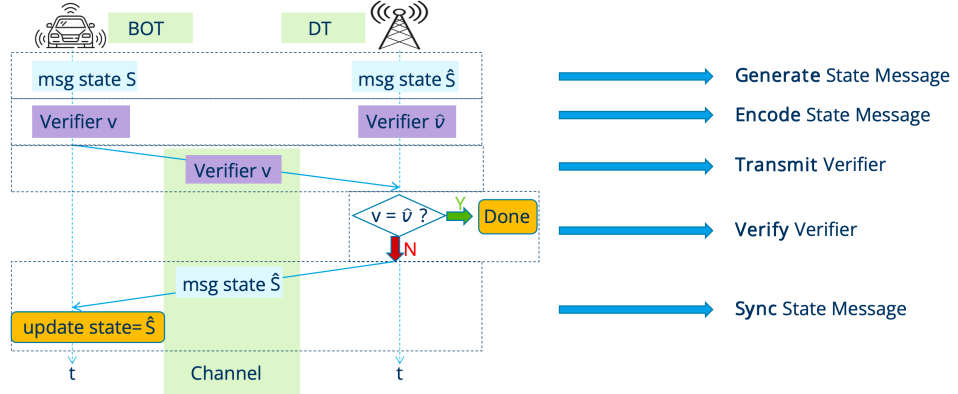


Figure 3.2: Data flow of ID Synchronization: High-bandwidth transmission is replaced by local encoding and conditional fallback.

3.2 Encoding Mechanisms and Reliability Analysis

The safety of the ID flow depends on the reliability of the Verify stage. A critical failure occurs if the states differ ($S_t \neq \hat{S}_t$) but their verifiers collide ($v = \hat{v}$), defined as a Type II Error (p_{miss}). We investigate two distinct encoding mechanisms, detailing their construction and theoretical error bounds.

3.2.1 Algebraic Implementation and Error Bound (RS-ID)

To integrate the RS-ID scheme into our synchronization protocol, we map the high-dimensional state vector S of length n_{data} to a polynomial over the finite field $\mathbb{F}_q$, where $q = 2^{n_{ver}}$.

Following the construction detailed in Section 2.3.2, the bitstream S is sliced into k coefficients. For the specific bandwidth constraints of our system model, the number of coefficients is determined by:

$$k = \left\lceil \frac{n_{data}}{n_{ver}} \right\rceil \quad (3.1)$$

The system generates the verifier v by evaluating the polynomial $P_S(x) = \sum_{i=0}^{k-1} s_i x^i$ at the shared seed r .

For the experimental evaluation and latency modeling, we apply the theoretical upper bound derived in Chapter 2 (Schwartz-Zippel Lemma). The probability of a Type II error (collision) is strictly modeled as:

$$P_{miss}^{RS} \leq \frac{k}{q} = \frac{\lceil n_{data}/n_{ver} \rceil}{2^{n_{ver}}} \quad (3.2)$$

This equation serves as the reliability constraint for the design: to ensure robust identification ($P_{miss} \ll 1$), the verifier length n_{ver} is selected such that the field size exponentially exceeds the data complexity k .

3.2.2 Cryptographic Construction and Error Analysis (SHA-256)

Unlike the algebraic structure of RS-ID, the security of hash-based identification relies on computational hardness assumptions. We verify the state using a truncated cryptographic hash function (SHA-256):

$$v = \text{Truncate}(\text{SHA-256}(S||r), n_{ver}) \quad (3.3)$$

where $||$ denotes concatenation with the random seed r .

To analyze the collision probability theoretically, we adopt the [Random Oracle Model \(ROM\)](#) paradigm introduced by Bellare and Rogaway [6]. In this model, the hash function $H(\cdot)$ is idealized as a mathematical oracle that maps every unique input query to a response selected uniformly and independently from the output domain $\{0, 1\}^{n_{ver}}$.

Under the [ROM](#), for any robot state S , the hash output $H(S||r)$ is a random variable following a uniform distribution over the verifier space of size $N = 2^{n_{ver}}$. Given a pre-calculated verifier

$\hat{v}$ from the Digital Twin, the probability of a collision (Type II error) is equivalent to the probability that a uniformly selected random value hits a specific target in the sample space.

$$P_{miss}^{SHA} = \Pr[H(S||r) = \hat{v}] = \frac{1}{|\{0, 1\}^{n_{ver}}|} = \frac{1}{2^{n_{ver}}} \quad (3.4)$$

This derivation relies on the property that in a uniform distribution, the probability of any single outcome is the reciprocal of the sample space size.

Comparing Equation (3.2) with Equation (3.4) highlights a key structural difference. In **RS-ID**, the error probability scales linearly with the data size ($k \approx n_{data}/n_{ver}$), reflecting the geometric fact that higher-degree polynomials admit more roots. Under the **ROM** assumption, the **SHA-256** collision probability is invariant to n_{data} . As a result, the cryptographic approach remains well-behaved for large state vectors (e.g., **Light Detection and Ranging (LiDAR)** point clouds), but it replaces the unconditional Schwartz–Zippel guarantee with a computational idealization.

3.3 End-to-End Latency Model

Based on the data flows defined in Section 3.1, we mathematically formulate the expected latency.

Since the traditional flow always executes the full transmission, its latency is deterministic:

$$T_{trad} = \frac{n_{data}}{B} \quad (3.5)$$

where B is the channel bandwidth.

In contrast, the ID flow is probabilistic. The total time includes the mandatory encoding and verifier transmission, plus the conditional penalty for fallback. Matching the stages in Figure 3.2:

$$T_{ID} = \underbrace{t_{ver}}_{\text{Encode}} + \underbrace{\frac{n_{ver}}{B}}_{\text{Transmit Verifier}} + \underbrace{p_{desync}(1 - p_{miss})\frac{n_{data}}{B}}_{\text{Sync (Fallback Penalty)}} \quad (3.6)$$

Here, t_{ver} denotes the computation time of the Encode stage, p_{desync} is the probability that the DT prediction fails ($S_t \neq \hat{S}_t$), and the fallback penalty $(1 - p_{miss})\frac{n_{data}}{B}$ is incurred only when a desynchronization exists and is successfully detected by the verifier.

This model highlights the fundamental trade-off: ID reduces the transmission term from $\frac{n_{data}}{B}$ to $\frac{n_{ver}}{B}$ but adds a computational cost t_{ver} . Therefore, ID is advantageous only when the bandwidth B is limited (making transmission expensive) or when the prediction accuracy is high (keeping p_{desync} small).

3.4 Tolerance-Based Verification (K-ID)

The ID data flow described in Section 3.1 assumes a strict equality check between a single predicted state $\hat{S}_t$ and the true state S_t . In practice, however, many cyber-physical systems can tolerate bounded deviations without requiring a full resynchronization. We therefore extend

the verification stage to a tolerance-based rule, widely referred to as **K-ID** in information theory literature [1].

Figure 3.3 illustrates the resulting protocol. The operational stages mirror the standard ID flow, but the decision criterion at the receiver is generalized: instead of checking equality against a single predicted verifier $\hat{v}$, the DT accepts if the received verifier matches any candidate in a precomputed set corresponding to the tolerable neighborhood of $\hat{S}_t$.

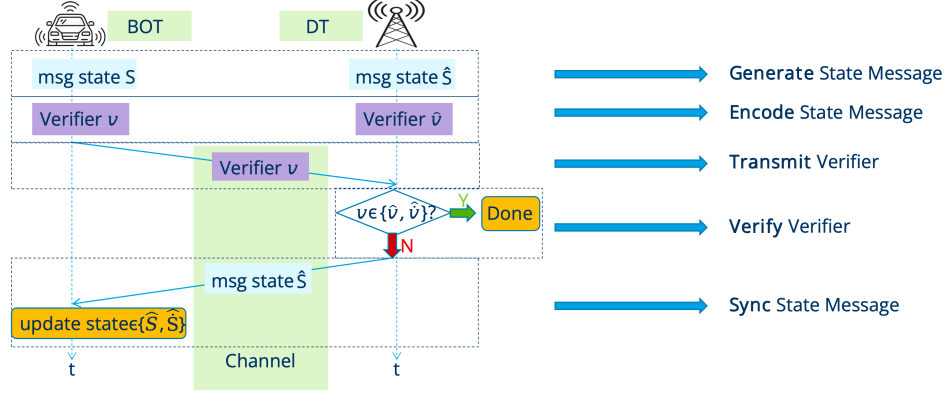


Figure 3.3: Data flow of K-ID Synchronization: High-bandwidth transmission is replaced by local encoding and conditional fallback.

3.4.1 Acceptance Set and Quantization

Let $\mathcal{Q}$ denote the domain of quantized states achievable by the robot, with a discretization resolution Δ . Given the DT prediction $\hat{S}_t$ and a physical tolerance radius r , the DT defines a semantic acceptance set $\mathcal{A}(\hat{S}_t)$ containing all quantized states within the radius:

$$\mathcal{A}(\hat{S}_t) = \{s \in \mathcal{Q} \mid \|s - \hat{S}_t\| \leq r\}. \quad (3.7)$$

The cardinality of this set is denoted by $K = |\mathcal{A}(\hat{S}_t)|$. Note that in this thesis, we refer to the scheme as K-ID following standard terminology [1], where K represents the size of the hypothesis set.

The Robot computes the verifier v using the selected encoding scheme $Enc(\cdot)$ on the current quantized state S_t . Simultaneously, the DT pre-computes the set of valid verifiers $\mathcal{V}_{set}$ corresponding to all candidates in $\mathcal{A}(\hat{S}_t)$:

$$\mathcal{V}_{set} = \{Enc(s) \mid s \in \mathcal{A}(\hat{S}_t)\}. \quad (3.8)$$

The specific construction of the encoding function $Enc(s)$ depends on the chosen algorithm. For SHA-256, we define $Enc(s) = \text{Truncate}(\text{SHA256}(s), n_{ver})$. For RS-ID, we define $Enc(s) = M_s(\beta)$, where $M_s(x)$ is the polynomial representation of state s and $\beta \in \mathbb{F}_{2^{n_{ver}}}$ is the designated evaluation point (tag position).

The DT accepts if the received verifier $v \in \mathcal{V}_{set}$. This implies the robot is in a "safe" state consistent with the tolerance region. Otherwise, it triggers fallback.

3.4.2 Collision Probability Analysis

Define an unsafe event as the true state lying outside the tolerance region, i.e., $S_t \notin \mathcal{A}(\hat{S}_t)$. A miss (false negative) occurs when the verifier $v(S_t)$ of an unsafe state coincidentally falls into the set of valid verifiers $\mathcal{V}_{set}$. The miss probability is bounded differently depending on the encoding mechanism.

Case 1: Cryptographic Hashing (SHA-256)

Under the Random Oracle assumption, the verifier for an unknown unsafe state is uniformly distributed over the code space $\{0, 1\}^{n_{ver}}$. By the union bound, the miss probability is simply the fraction of the space covered by the K valid candidates:

$$p_{miss}^{SHA} \leq \sum_{s \in \mathcal{A}(\hat{S}_t)} \Pr[v(S_t) = v(s)] \approx \frac{K}{2^{n_{ver}}}. \quad (3.9)$$

Crucially, this bound is independent of the data payload size n_{data} , making it scalable for large state vectors.

Case 2: Algebraic Coding (RS-ID)

For RS-ID, the verification relies on polynomial evaluation over a finite field $\mathbb{F}_q$, where $q = 2^{n_{ver}}$. As established in Section 2.3.2, two distinct messages of length n_{data} (interpreted as polynomials of degree $< n_{data}$) collide in at most $n_{data} - 1$ points. When checking against a set of K valid candidates, the worst-case collision probability is the sum of collision probabilities with each candidate:

$$p_{miss}^{RSID} \leq \sum_{s \in \mathcal{A}(\hat{S}_t)} \frac{n_{data} - 1}{q} = K \cdot \frac{n_{data} - 1}{2^{n_{ver}}}. \quad (3.10)$$

Comparing Eq. (3.9) and Eq. (3.10), we observe that RS-ID introduces a penalty factor proportional to the payload size (n_{data}). While RS-ID provides a deterministic bound, this bound loosens linearly with data size, making it less efficient than SHA-256 for K-ID when n_{data} is large, unless the field size n_{ver} is increased strictly.

3.4.3 Latency and Safety Analysis

To connect the tolerance rule to system performance, we define the unsafe probability $p_{unsafe}(r)$ as the likelihood that the true state S_t falls outside the acceptance radius r . Assuming the prediction error $e = S_t - \hat{S}_t$ follows a Zero-Mean Gaussian distribution $\mathcal{N}(0, \sigma^2)$, this probability is:

$$p_{unsafe}(r) = \Pr[|e| > r] = 2 \left(1 - \Phi \left(\frac{r}{\sigma} \right) \right). \quad (3.11)$$

where $\Phi(\cdot)$ is the CDF of the standard normal distribution.

A fallback (full transmission) is triggered only when the Digital Twin correctly detects an unsafe state. This occurs when the state is unsafe AND the hash verification does not generate a false positive (miss). The probability of fallback is therefore:

$$p_{fallback} = p_{unsafe}(r) \cdot (1 - p_{miss}). \quad (3.12)$$

The expected end-to-end latency T_{KID} is derived by substituting this fallback rate into the general latency structure defined in Section 3.3:

$$T_{KID} = t_{ver} + \frac{n_{ver}}{B} + p_{unsafe}(r)(1 - p_{miss})\frac{n_{data}}{B}. \quad (3.13)$$

This analytical model elucidates the central trade-off explored in this thesis: increasing the tolerance radius r reduces $p_{unsafe}(r)$ (which lowers latency by reducing fallbacks), but simultaneously enlarges the candidate set size K , which increases p_{miss} (compromising safety). The empirical parameter sweep in Chapter 5 instantiates this model to visualize the safety-latency optimization surface.

3.4.4 Sensitivity Analysis of Quantization Resolution

While the tolerance parameter K determines the physical scope of the safety zone, the granularity of this coverage is governed by the quantization resolution Δ . The choice of Δ imposes a fundamental trade-off between physical control fidelity and decoder computational scalability.

The primary determinant of the decoder’s computational workload is the cardinality of the acceptance set $|\mathcal{A}_K|$, which dictates the number of hash evaluations required per prediction step. For a fixed physical tolerance radius R_{phy} —such as the “Safe Harbor” radius required to encapsulate bounded noise—the set size scales inversely with the resolution:

$$|\mathcal{A}_K| \approx \frac{2R_{phy}}{\Delta} \quad (3.14)$$

This hyperbolic relationship implies that minimizing quantization error by driving $\Delta \rightarrow 0$ inherently leads to an explosion in the computational burden on the Digital Twin. As the resolution becomes finer, the linear increase in fidelity incurs a non-linear penalty in processing overhead, potentially creating a bottleneck for high-frequency control loops.

Consequently, the evaluation of system safety must extend beyond probabilistic collision risk to encompass computational feasibility. We formulate the optimization of Δ as a constraint satisfaction problem rather than a simple minimization of error. The objective is to identify the finest resolution Δ_{min} such that the worst-case decoder latency $T_{dec}(R_{max}, \Delta)$ remains bounded by the strict real-time control budget T_{budget} (e.g., < 1.5 ms). This framework allows us to identify the operational limit that maximizes state precision without saturating the edge processor’s computational resources.

3.5 Dynamic Evaluation Model: Random Walk with Drift

The static state model utilized in the parameter sweep (assuming i.i.d. samples of S_t) effectively estimates per-tick miss probabilities; however, it fails to capture error accumulation over time. To investigate continuous unsafe operation and boundary breaches resulting from missed detections, we introduce a time-correlated dynamic model wherein the robot drifts stochastically while the DT maintains its last validated state via a zero-order hold.

We model the scalar robot state evolution as a discrete-time random walk:

$$S_t = S_{t-1} + \delta_t, \quad (3.15)$$

where δ_t denotes the process noise. We consider two noise profiles: a uniform increment $\delta_t \sim \mathcal{U}(-\Delta_{step}, \Delta_{step})$ representing hard kinematic constraints, and a Gaussian increment $\delta_t \sim \mathcal{N}(0, \sigma_{step}^2)$ capturing unbounded disturbances.

On the receiver side, the **DT** employs a **Zero-Order Hold (ZOH)** predictor logic. In the absence of an explicit correction packet, it assumes the synchronization is maintained:

$$\hat{S}_t = \hat{S}_{t-1} \quad \text{unless a fallback correction is triggered.} \quad (3.16)$$

Consequently, the instantaneous drift error $E_t = |S_t - \hat{S}_t|$ grows monotonically between corrections.

At each simulation tick, the protocol enforces the **K-ID** tolerance gate:

$$\begin{cases} \text{if } E_t \leq K : & \text{Silent (masked drift),} \\ \text{if } E_t > K : & \text{Transmission and Verification.} \end{cases} \quad (3.17)$$

When $E_t > K$, the Robot transmits a verifier. The **DT** checks this verifier against its local estimate $\hat{S}_t$. If a hash collision occurs (with probability $p_{miss,K}$), the system suffers a safety miss: the **DT** erroneously accepts the outdated $\hat{S}_t$, allowing the drift to accumulate further. Conversely, if no collision occurs, the **DT** detects the discrepancy, triggers a fallback, and resets $\hat{S}_t \leftarrow S_t$.

To align the dynamic simulator with the Random Oracle analysis in Section 3.4, we model the probability of collision during verification events (i.e., ticks where $E_t > K$) using the acceptance-set proxy:

$$p_{miss,K}^{SHA} \approx \frac{|\mathcal{A}_K|}{2^{n_{ver}}}. \quad (3.18)$$

Here, $|\mathcal{A}_K| \approx 2\lfloor K/\Delta \rfloor + 1$ represents the cardinality of the acceptance set over the state domain quantized with resolution Δ .

4 Implementation

This chapter summarizes the practical implementation used to instantiate and evaluate the theoretical models introduced in Chapter 3. The evaluation is organized around three complementary models. First, we implement the end-to-end ID latency model and generate bandwidth and desynchronization sensitivity plots. Second, we implement the tolerance-based K-ID protocol and perform a parameter sweep over (K, n_{ver}) to expose the safety-performance trade-off and identify Pareto-optimal operating points. Third, we implement a time-correlated dynamic model in which K-ID operates under random-walk drift, allowing us to study accumulation effects such as sustained boundary breaches after missed corrections.

4.1 Data Packet Structure Abstraction

While the analysis in Chapter 3 models transmission cost in terms of n_{ver} and n_{data} , practical deployments encapsulate these fields in a communication frame. Figure 4.1 illustrates the abstract frame layout assumed in this thesis. In normal operation, the Robot transmits only a compact verifier; when verification fails, the protocol falls back to transmitting the full state payload. The K-ID protocol uses the same message structure as ID; it differs only in the receiver-side acceptance rule (set membership rather than strict equality).

In the latency model, the header, timestamp, and checksum/CRC are treated as fixed overheads and are omitted for clarity. If required, these can be incorporated by adding a constant number of bits to both the verifier-only and fallback frames.

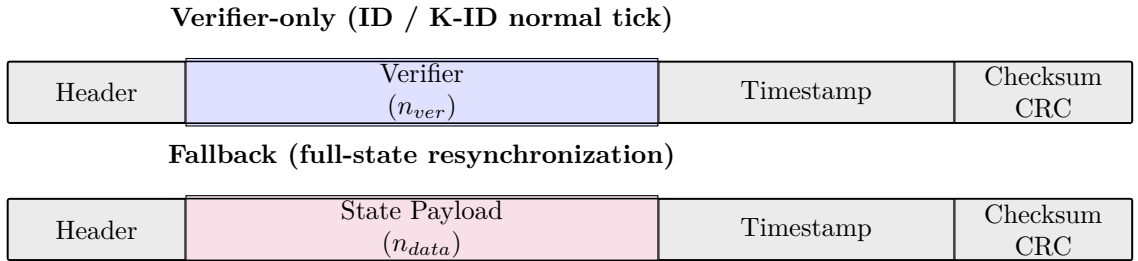


Figure 4.1: Abstract ID message format used for modeling. The normal path transmits only a verifier, while the fallback path transmits the full state payload. Fixed overhead fields are shown for completeness but are not explicitly modeled in the latency equations.

4.2 Software Architecture

To facilitate rapid prototyping and evaluation of the proposed models, we implement a modular Python simulator.

Figure 4.2 shows the high-level decomposition used throughout the Python experiments. The simulator is structured around three interchangeable components—Encoder, Channel/Dynamics, and Decoder/Decision. This separation allows swapping SHA-256 and RS-ID encoders, switching between strict ID and tolerance-based K-ID decision rules, and evaluating both i.i.d. and time-correlated models without entangling implementation details.

The interface between Encoder and Channel is a transmitted packet/frame. In normal operation this contains only a verifier tag; on fallback it contains the full state payload. This matches the message abstraction in Figure 4.1.

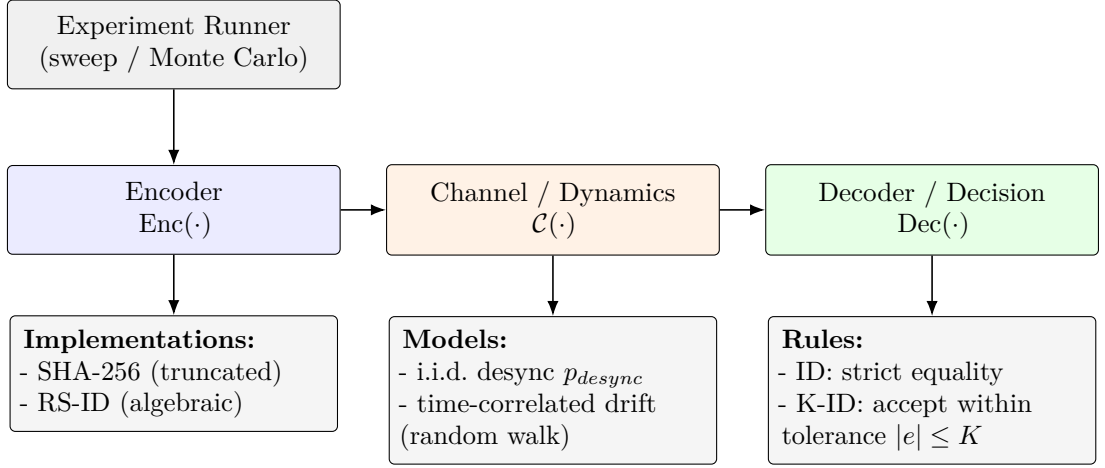


Figure 4.2: High-level simulator architecture used in the Python experiments. Encoder, Channel/Dynamics, and Decoder are decoupled so that coding schemes (SHA-256 vs. RS-ID), acceptance rules (ID vs. K-ID), and stochastic models (i.i.d. vs. time-correlated drift) can be varied independently.

4.3 Latency Plot Generation

To quantify the implications of the hybrid latency model, we evaluate the expected latency under systematic sweeps of the channel bandwidth B and the desynchronization probability p_{desync} . All plots are generated for multiple payload sizes n_{data} , verifier lengths n_{ver} , and for both encoding mechanisms (SHA-256 vs. RS-ID), allowing us to compare regimes where ID provides a net latency advantage over traditional transmission.

The latency evaluation uses the parameter grid defined the same as [14], we use a baseline desynchronization probability of $p_{desync} = 0.1$; this value is held fixed for the bandwidth sweep and is varied explicitly when evaluating desynchronization sensitivity. We evaluate two data payloads: a small 96-bit message and a larger 4001-bit message. For each payload, we test verifier lengths of $n_{ver} \in \{4, 16\}$ bits, representing a small and a large verifier size. The compute overheads t_{ver} are taken from empirical measurements (Table 5.1), and the collision probabilities p_{miss} are computed from the theoretical bounds (Table 5.2). All t_{ver}

measurements report mean $\pm$ 95% CI half-width, and the timing configuration is chosen to keep the relative CI within approximately 5% for each point. The scripts used for these measurements are located in `experiments/`:

4.3.1 Logic

For each configuration (n_{data}, n_{ver}) and each system (SHA-256 vs. RS-ID), the script evaluates the expected latency

$$L_{ID}(B) = t_{ver} + \frac{n_{ver}}{B} + p_{desync} (1 - p_{miss}) \frac{n_{data}}{B} \quad (4.1)$$

over a log-spaced bandwidth grid $B \in [B_{min}, B_{max}]$. The traditional baseline is computed as $L_{trad}(B) = n_{data}/B$. The resulting curves are plotted on log-log axes to reveal the break-even regime where ID-based synchronization becomes beneficial.

The generated figures are shown in Chapter 5 for $n_{data} = 96$ bits (Figure 5.1a) and for $n_{data} = 4001$ bits (Figure 5.1b).

In addition, we visualize the sensitivity to desynchronization by sweeping $p_{desync} \in [0, 1]$ at a fixed bandwidth and plotting L_{ID} against p_{desync} for the same parameter grid (Figures 5.2a and 5.2b).

4.3.2 SHA-256

The collision probability for a truncated hash function under the Random Oracle Model is independent of the data size and is given by Equation 3.9:

$$P_{miss}^{SHA} = 2^{-n_{ver}} \quad (4.2)$$

The corresponding numerical values for our selected verifier lengths are reported in Table 5.2 in Chapter 5.

4.3.3 RS-ID

The collision probability for the algebraic construction is bounded by the Schwartz-Zippel Lemma, as shown in Equation 3.2. This bound depends on both data size (n_{data}) and verifier length (n_{ver}).

$$P_{miss}^{RS} \leq \frac{\lceil n_{data, bits} / n_{ver} \rceil}{2^{n_{ver}}} \quad (4.3)$$

The computed values for $n_{data} \in \{96, 4001\}$ bits and $n_{ver} \in \{4, 16\}$ bits (with the bound capped at 1 when applicable) are reported in Table 5.2 in Chapter 5. These values highlight how the theoretical safety of RS-ID degrades with larger data sizes for a fixed verifier length, a key distinction from the cryptographic approach.

4.3.4 LiDAR-Scale Scalability Parameters

To study how the compute overhead t_{ver} scales with high-dimensional sensor payloads, we perform an additional scalability experiment at $n_{ver} = 16$. We measure verifier generation time for payload sizes representative of LiDAR message chunks and frame-scale payloads:

$$n_{data} \in \{12 \text{ B}, 501 \text{ B}, 100 \text{ KB}, 1 \text{ MiB}, 5 \text{ MiB}\} \quad (4.4)$$

where 12 B and 501 B correspond to the 96-bit and 4001-bit baselines used elsewhere in the thesis. For [SHA-256](#), the payload is represented as a byte list of length $\lceil n_{data}/8 \rceil$. For [RS-ID](#), the payload is modeled as packed $\text{GF}(2^{n_{ver}})$ symbols with

$$k = \left\lceil \frac{n_{data}}{n_{ver}} \right\rceil \quad (4.5)$$

so that increasing payload size increases the number of field coefficients processed by the algebraic encoder.

4.3.5 Evaluation Scenarios (Bandwidth Baselines)

We use two bandwidth operating points, each tailored to the objective of the corresponding evaluation. For the baseline [ID](#) sensitivity results in Section 5.2 (Chapter 5), we report single-bandwidth slices at $B = 5$ Mbps to isolate the impact of prediction quality (p_{desync}) and payload size (n_{data}) without making the channel the sole bottleneck; this also complements the full bandwidth sweeps (10 kbps–10 Mbps). For the [K-ID](#) parameter sweep and Pareto analysis in Section 5.3, we adopt a bandwidth-constrained setting ($B = 1$ Mbps) because [K-ID](#) primarily trades computation for reduced traffic (fewer fallback transmissions), and this benefit is most visible when the transmission term n_{data}/B dominates.

4.4 K-ID Simulation Setup and Implementation

To validate the theoretical K-ID framework proposed in Chapter 3, we implemented a stochastic simulation environment that quantifies the trade-off between the semantic tolerance radius K and safety risks. The simulation mimics a constrained edge-link scenario, bridging the gap between theoretical error bounds and practical system performance.

4.4.1 Error Models and State Generation

The simulation models the robot’s state S_t as a continuous random variable constrained to the physical domain of $[0, 100]$. To strictly evaluate the impact of error distribution shapes, we implemented two contrasting noise models—Gaussian and Uniform—while controlling for their noise energy.

For the baseline Gaussian model ($S_t \sim \mathcal{N}(\mu, \sigma^2)$), we centered the mean at $\mu = 50$ and specifically selected a standard deviation of $\sigma = 16.0$. This parameter choice is engineered to maximize the utilization of the physical state space without violating its boundaries. Since the effective width of the Gaussian distribution (3σ) is 48, the probability mass is concentrated within the interval $[2, 98]$. This configuration ensures that the simulation captures extreme,

high-magnitude deviations typical of unbounded noise, while preserving the mathematical integrity of the Gaussian profile by avoiding significant truncation (clipping) at the physical limits of 0 and 100.

For the comparative Uniform model ($S_t \sim \mathcal{U}[\mu - a, \mu + a]$), we enforce a strict variance-matching constraint to isolate the effect of bounded support. Equating the variance of the Uniform distribution ($a^2/3$) to the Gaussian baseline ($\sigma^2 = 16^2$) yields a half-width of $a = 16\sqrt{3} \approx 27.71$. Consequently, the Uniform model is defined over the interval $[22.29, 77.71]$. This derivation establishes a critical theoretical threshold: any tolerance radius K exceeding 27.71 should theoretically eliminate all collision risks for the Uniform case, a hypothesis we explicitly test in the parameter sweep.

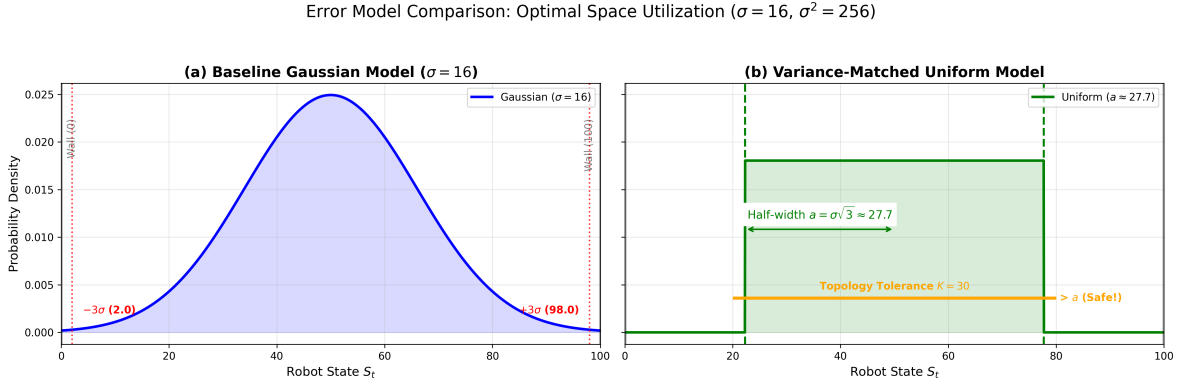


Figure 4.3: Visual comparison of the variance-matched error models configured with $\sigma = 16$. (a) The Gaussian baseline is engineered to maximize physical space utilization: the effective 3σ range (48) fits tightly within the physical bounds ($50 \pm 48 = [2, 98]$), preventing truncation artifacts while maintaining high noise energy. (b) The Uniform model ($a \approx 27.71$) illustrates the “Safe Harbor” principle: the selected tolerance radius $K = 30$ (orange bar) explicitly exceeds the noise support a , theoretically guaranteeing zero collision risk, whereas the Gaussian model retains residual tail risks outside the tolerance zone.

4.4.2 Computational Scalability and Latency Model

To quantify the computational scalability of the K-ID decoder, we conduct a fine-grained sensitivity analysis by sweeping the quantization resolution Δ from 1.0 down to 0.01 on a logarithmic scale. We evaluate the latency impact across a spectrum of physical safety requirements, ranging from high-precision control ($R = 2$) and intermediate tolerance ($R = 10$) to the maximum “Safe Harbor” configuration ($R = 30$) required to encapsulate the Uniform noise boundary ($a \approx 27.71$).

The total decoder latency is modeled linearly as the product of the acceptance set cardinality $|\mathcal{A}_K|$ and the unit verification overhead t_{ver} . Crucially, this estimation assumes sequential execution—implying no parallelization or hardware acceleration—on the Digital Twin side. By adopting this serial processing assumption, we establish a conservative, worst-case baseline for performance evaluation. While parallel implementation strategies (e.g., SIMD instructions or GPU offloading) could further reduce processing time, demonstrating feasibility under strict

serial constraints confirms the algorithm’s fundamental efficiency even on resource-limited computing nodes.

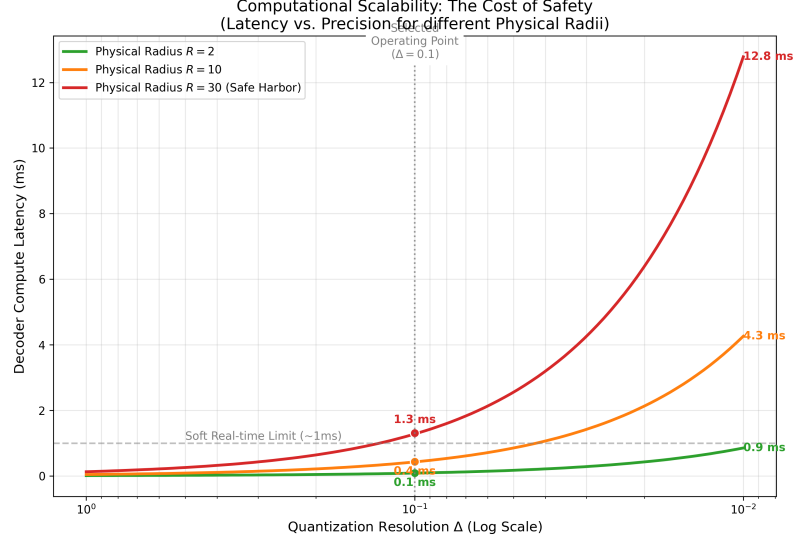


Figure 4.4: Computational scalability analysis showing decoder latency as a function of quantization resolution Δ . The red curve represents the “Safe Harbor” configuration ($R = 30$). At the high-precision limit ($\Delta = 0.01$), latency reaches a prohibitive 12.8 ms. However, at our selected operating point ($\Delta = 0.1$, vertical dotted line), the latency drops to ≈ 1.3 ms, balancing maximum physical coverage with real-time computational feasibility.

The results of this sensitivity sweep, illustrated in Figure 4.4, reveal a critical scalability bottleneck governed by the choice of resolution. As the quantization Δ becomes finer, the computational cost grows hyperbolically, a trend that is most pronounced for the large-tolerance “Safe Harbor” configuration ($R = 30$, depicted by the red curve). When applying a soft real-time constraint of approximately 1 ms, the data demonstrates that a high-precision setting of $\Delta = 0.01$ is operationally infeasible; it drives the decoder latency to 12.8 ms, violating the control budget by an order of magnitude.

In contrast, relaxing the resolution to $\Delta = 0.1$ drastically reduces the computational burden. At this operating point, the latency for the worst-case $R = 30$ scenario drops to approximately 1.3 ms. Although this marginally exceeds the soft target, it remains within a manageable envelope for real-time systems, whereas smaller safety radii ($R = 2, 10$) perform well below 0.5 ms. Consequently, we select $\Delta = 0.1$ as the system’s fixed operating resolution. This choice represents the engineering limit of precision, providing the finest granularity that permits full “Safe Harbor” coverage without causing a computational breakdown in the Digital Twin.

4.4.3 System Configuration and Parameter Sweep

The simulation assumes a payload ($n_{data} = 4001$ bits) transmitted over a 1 Mbps link. The quantization resolution is fixed at $\Delta = 0.1$ for the primary evaluation.

Instead of sweeping the abstract integer tolerance K , we sweep the Physical Tolerance Radius R to represent concrete safety requirements. The parameter grid is defined as:

$$R \in \{2, 5, 10, 20, 30\} \quad (\text{Physical Units}) \quad (4.6)$$

The corresponding integer tolerance steps for the K-ID algorithm are derived dynamically as $K = \lceil R/\Delta \rceil$. For the baseline $\Delta = 0.1$, this results in an effective K sweep of $\{20, 50, 100, 200, 300\}$.

The inclusion of $R = 30$ is motivated by the variance-matched Uniform noise boundary $a \approx 27.71$. In this setting, $R = 30$ constitutes a “Safe Harbor” configuration ($R > a$), which should eliminate unsafe events under the bounded model. In contrast, for the Gaussian model with unbounded support, the same radius ($R = 30 \approx 1.875\sigma$) is expected to retain a non-zero tail probability of excursions beyond the tolerance. This bounded-vs.-unbounded contrast is used explicitly in Chapter 5.

4.4.4 Visualization Methodology

To comprehensively analyze the trade-off space, we employ a dual visualization strategy combining parameter grid heatmaps with Pareto efficiency analysis.

First, the raw simulation results are mapped onto a two-dimensional grid defined by the verifier length n_{ver} and the physical tolerance radius R . Within each grid cell (R, n_{ver}) , we visualize two distinct scalar fields: the empirical speedup factor, displayed on a linear color scale to highlight proportional bandwidth savings, and the overall risk probability $P(\text{unsafe} \cap \text{accept})$, displayed on a logarithmic scale to differentiate between rare safety events and high-probability failures. Numeric values are explicitly annotated within each cell to facilitate precise quantitative comparison.

Second, to identify the optimal design configurations, we project these grid points onto a unified Risk-vs-Speedup scatter plot. By analyzing the distribution of performance metrics, we extract the Pareto Frontier—the set of non-dominated configurations where no other parameter combination yields a higher speedup for the same or lower level of risk. This frontier visually demarcates the theoretical efficiency limit of the K-ID protocol, thereby guiding the selection of the “sweet spot” parameters that maximize compression efficiency while satisfying strict safety constraints.

4.5 Implementation and Experimental Design

To rigorously evaluate the K-ID protocol under dynamic conditions, we developed a discrete-time simulation framework comprising three distinct modules: a stochastic robot agent, a bandwidth-constrained channel, and a Digital Twin (DT) receiver. This section details the modeling choices, the protocol logic implementation, and the rationale behind the parameter configurations used to generate the subsequent results.

4.5.1 Dynamic Error Evolution and Protocol Logic

Unlike static error checks, dynamic evaluation requires modeling the time-correlated accumulation of drift. We implement the robot’s state evolution S_t as a standard diffusive process, specifically a Gaussian Random Walk defined by $S_t = S_{t-1} + \delta_t$, where the increment δ_t follows a normal distribution $\mathcal{N}(0, \sigma_{step}^2)$. For all experiments, we standardize on a unit step deviation ($\sigma_{step} = 1.0$). This choice avoids arbitrary parameter tuning, as it represents a canonical diffusion process where the variance of the uncorrected state grows linearly with time, providing a robust and reproducible challenge for the synchronization protocol.

The K-ID protocol is implemented as a strict guardrail mechanism superimposed on this physical state evolution. The decision logic at each time tick proceeds sequentially. First, the system performs a tolerance check by calculating the absolute deviation $E_t = |S_t - \hat{S}_{t-1}|$ between the true state and the DT’s estimate. If this deviation remains within the radius R , the protocol enforces silence to suppress bandwidth usage. Conversely, when the deviation exceeds the tolerance ($E_t > R$), a verification event is triggered. To simulate the cryptographic behavior efficiently over millions of simulation ticks, we abstract the hash function using a probabilistic collision model. Rather than computing raw SHA-256 hashes at every step, the verification result is modeled as a Bernoulli trial where the probability of a “safety miss” (collision) is calculated dynamically based on the acceptance set size ratio $|\mathcal{A}_K|/2^{n_{ver}}$. This abstraction decouples the logical correctness of the protocol from the specific choice of hash function, focusing the analysis on the theoretical entropy bounds.

4.5.2 Experimental Strategy: Stress Test and Parameter Sweep

Our evaluation strategy consists of two complementary phases: a targeted stress test for visualization and a comprehensive parameter sweep for statistical quantification.

For the qualitative Stress Test, we explicitly designed a configuration intended to induce and visualize rare failure modes within an observable simulation window. We strictly adhere to the optimal quantization resolution ($\Delta = 0.1$) derived from the static analysis. This is paired with an intermediate physical tolerance ($R = 10$) and, crucially, a 12-bit verifier ($n_{ver} = 12$). With an acceptance set of $|\mathcal{A}_K| \approx 201$, a 12-bit space yields a collision probability of approximately 4.9%. This intentionally elevated risk level serves as a heuristic device to amplify the frequency of safety misses, allowing us to capture the anatomy of a failure event—specifically, a drift break-out masked by a hash collision—without requiring computationally prohibitive simulation durations.

For the quantitative Parameter Sweep, we expand the evaluation to map the broader risk landscape. While fixing the drift dynamics ($\sigma = 1.0, \Delta = 0.1$), we systematically vary the tolerance radius $K \in \{2, 5, 10, 20\}$ and the verifier length $n_{ver} \in \{8, 12, 16\}$. This grid search allows us to decouple the “masking effect” of physical tolerance from the “collision risk” of cryptographic verification, providing the statistical basis for the trade-off analysis.

4.5.3 Metrics for Dynamic Safety

To quantify the trade-off between efficiency and risk, we define three primary metrics for the dynamic experiments. The first is the Inter-Fallback Time (IFT), defined as the duration between two successful correction updates. A larger mean IFT indicates higher channel

efficiency and reduced network contention, effectively measuring the protocol’s ability to compress traffic.

To measure safety risks, we define Breach Duration (L_K) as the number of consecutive ticks during which the system remains in an unsafe state ($|S_t - \hat{S}_t| > K$) due to hash collisions. This measures the temporal persistence of safety violations. Complementing this is the Maximum Drift, which records the peak absolute error reached during a simulation run. While Breach Duration quantifies how long the system is blind, Maximum Drift quantifies the worst-case physical magnitude of the error, serving as a proxy for potential physical damage in a real-world control scenario.

5 Results and Evaluation

This chapter presents the empirical evaluation of the proposed [ID](#)-based synchronization protocols. We first characterize the system’s fundamental parameters—computational overhead and collision probability—to calibrate the theoretical models. We then perform a sensitivity analysis to identify the bandwidth and prediction regimes where [ID](#) offers a latency advantage over traditional transmission. Next, we conduct a comprehensive parameter sweep of the [K-ID](#) protocol, quantifying the safety-performance trade-offs and identifying Pareto-optimal configurations for deployment. We conclude with a dynamic evaluation based on the random walk drift model to highlight time-correlated risk accumulation.

5.1 Empirical Parameter Verification

To instantiate the latency models derived in Chapter [3](#), we first experimentally verify the computational cost (t_{ver}) and the theoretical reliability limits (p_{miss}) of the encoding mechanisms.

5.1.1 Encoding Latency (t_{ver})

The encoding time t_{ver} sets the irreducible latency floor for [ID](#), even when bandwidth is effectively infinite. We measured t_{ver} over 100,000 operations for both [SHA-256](#) and [RS-ID](#). The results, summarized in Table [5.1](#), reveal a distinct structural divergence between the two approaches.

Table 5.1: Measured verification time t_{ver} (mean $\pm$ 95% CI half-width). **SHA-256** shows stable performance for fixed payloads, while **RS-ID** is sensitive to the verifier length due to field arithmetic complexity.

System (Payload)	Verifier Length (n_{ver})	
	4 bits	16 bits
<i>Cryptographic (SHA-256)</i>		
96 bits	$2.13 \pm 0.04 \mu s$	$2.24 \pm 0.01 \mu s$
4001 bits	$14.65 \pm 0.31 \mu s$	$15.41 \pm 0.09 \mu s$
<i>Algebraic (RS-ID)</i>		
96 bits	$14.55 \pm 0.39 \mu s$	$3.79 \pm 0.02 \mu s$
4001 bits	$51.78 \pm 0.94 \mu s$	$38.57 \pm 0.36 \mu s$

SHA-256 demonstrates robust stability; its runtime is dominated by the input block size and is effectively invariant to the output truncation length (n_{ver}). In contrast, **RS-ID** displays a counter-intuitive behavior where increasing the verifier length decreases latency (e.g., $14.55 \rightarrow 3.79 \mu s$). This is an artifact of the polynomial construction: a larger n_{ver} allows more bits to be packed into each field coefficient, reducing the total number of coefficients (k) required to represent the payload.

5.1.2 Theoretical Reliability Limits

Table 5.2 presents the theoretical collision probabilities calculated from the bounds in Chapter 3.

Table 5.2: Theoretical collision probability p_{miss} . **RS-ID** fails at short blocklengths ($n_{ver} = 4$) due to field saturation ($p_{miss} = 1.0$), whereas **SHA-256** maintains robust collision resistance.

System	Verifier Length (n_{ver})	
	4 bits	16 bits
SHA-256 (Random Oracle)	6.25×10^{-2}	1.53×10^{-5}
RS-ID (Schwartz-Zippel)	1.00 (<i>Saturated</i>)	9.16×10^{-5}

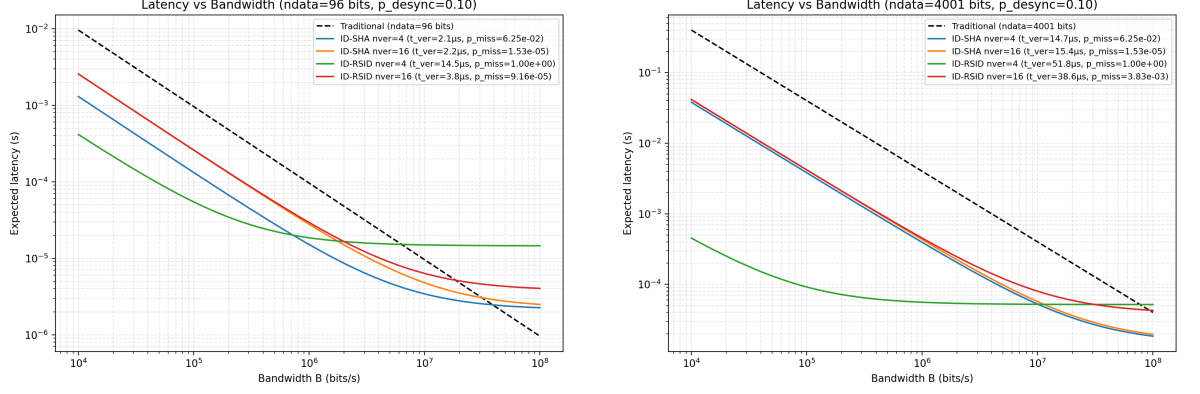
One critical finding is the failure of **RS-ID** at $n_{ver} = 4$. With a field size of only $q = 16$, the polynomial degree d for a 96-bit payload exceeds the field size ($d \geq q$), and the bound saturates at $p_{miss} = 1.0$. This confirms that algebraic codes require a minimum field headroom to function, establishing a hard lower bound on n_{ver} that does not exist for cryptographic hashes.

5.2 End-to-End Latency Evaluation

We now integrate the measured parameters into the hybrid latency model to evaluate system performance under varying network constraints.

5.2.1 The Break-Even Analysis (Bandwidth Sensitivity)

Figure 5.1 analyzes the expected latency across a bandwidth spectrum from 10 kbps to 10 Mbps.



(a) Small payload (96 bits): cross-over at ≈ 2 Mbps. (b) Large payload (4001 bits): ID advantage persists > 10 Mbps.

Figure 5.1: End-to-end latency vs. channel bandwidth. The plots reveal the break-even point where the computational cost of ID (t_{ver}) intersects with the transmission cost of the baseline. In the low-bandwidth regime ($B < 1$ Mbps), ID offers up to $10\times$ latency reduction.

The results illustrate the fundamental trade-off between computation and communication. In the transmission-bound regime ($B < 1$ Mbps), latency is dominated by the physical transmission time (T_{trans}), and ID outperforms the baseline by compressing the required transmission from n_{data} to n_{ver} . In the computation-bound regime ($B > 10$ Mbps), transmission becomes negligible and the latency floor is set by the encoding time t_{ver} ; since the traditional scheme has zero encoding overhead, it eventually becomes faster. For large payloads (Figure 5.1b), the break-even point shifts to higher bandwidths, indicating that ID-based synchronization becomes increasingly valuable as data complexity grows.

5.2.2 Operational Bounds (Prediction Sensitivity)

Figure 5.2 examines the impact of the DT's prediction quality (p_{desync}) at a fixed bandwidth of 5 Mbps. The intersection points in Figure 5.2 define the viability region of the protocol. For standard telemetry (96 bits), ID requires a prediction accuracy better than 20% ($p_{desync} < 0.8$) to yield net benefits. This indicates that ID remains practical in dynamic environments, provided the DT's kinematic model remains sufficiently coupled to reality.

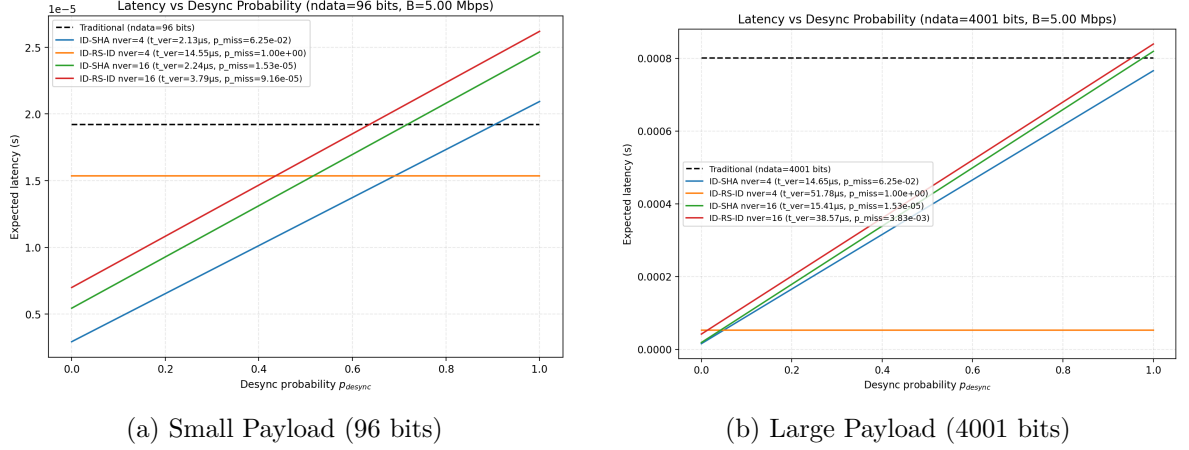


Figure 5.2: Latency sensitivity to prediction error. **ID** relies on accurate predictions. For small payloads (a), the system must maintain $p_{desync} < 0.8$ to outperform the baseline. For large payloads (b), the transmission savings are so significant that **ID** remains superior even with poor predictions.

5.2.3 LiDAR-Scale Payload Scalability

The baseline measurements above use small telemetry payloads (96 bits and 4001 bits). For high-dimensional sensors such as **LiDAR**, the payload size can grow by orders of magnitude. We therefore extend the t_{ver} benchmark to payloads up to 5 MiB (Section 4.3.4).

Figure 5.3 (visualized from Table 5.3) highlights a clear crossover in computational efficiency governed by payload size. For small payloads (< 100 B), **RS-ID** exhibits a slight performance edge (e.g., $1.81 \mu s$ vs. $2.06 \mu s$ at 96 bits), consistent with a small fixed overhead for polynomial evaluation while **SHA-256** incurs a startup cost due to message padding and the fixed compression routine. For large payloads (> 1 KiB), **SHA-256** significantly outperforms **RS-ID** (e.g., about $2\times$ faster at 5 MiB). While both approaches scale linearly in the number of processed bytes ($O(N)$), **SHA-256** relies primarily on bitwise operations that are heavily optimized on modern CPUs, whereas the current **RS-ID** implementation is dominated by memory-heavy finite-field multiplication table lookups.

Overall, for high-throughput sensor payloads (**LiDAR**/video), cryptographic hashing provides superior scalability in this prototype implementation. For distinct control signals and small telemetry payloads, both methods remain computationally negligible (on the order of a few microseconds).

Table 5.3: Measured verifier generation time t_{ver} for LiDAR-scale payload sizes ($n_{ver} = 16$). Values are mean $\pm$ 95% CI half-width; CI is controlled to $\leq 5\%$ relative error.

Payload	n_{data} (bits)	SHA-256 t_{ver} (μs)	RS-ID t_{ver} (μs)
12 B	96	2.06 ± 0.06	1.81 ± 0.02
501 B	4001	15.27 ± 0.30	24.69 ± 0.13
100 KB	819200	2702.45 ± 20.69	5208.16 ± 139.68
1 MiB	8388608	28388.94 ± 916.15	52728.32 ± 2396.59
5 MiB	41943040	140418.57 ± 1286.59	271339.25 ± 7741.95

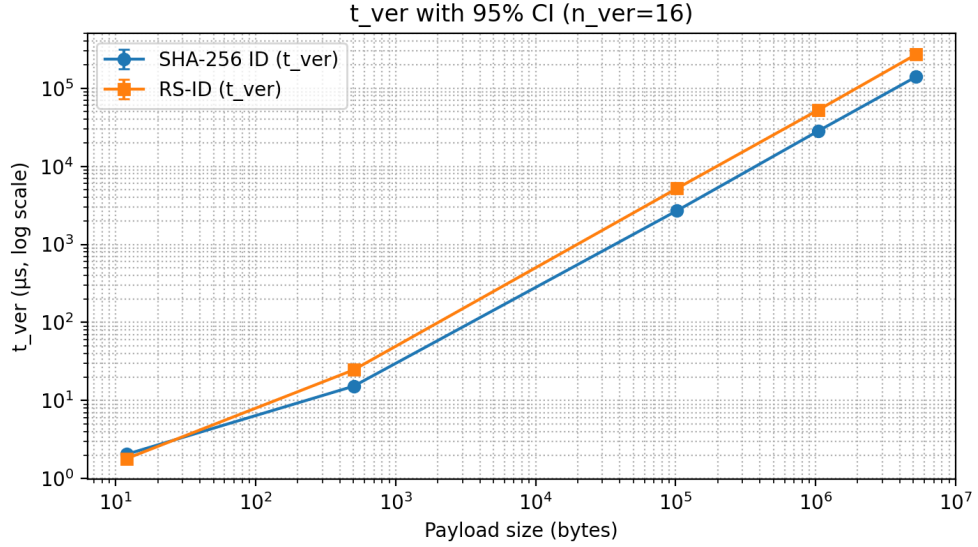


Figure 5.3: t_{ver} scalability vs. payload size at $n_{ver} = 16$ (log-log). Error bars show 95% CI half-width with relative CI controlled to $\leq 5\%$.

5.2.4 Overall Latency vs. Payload Size (Fixed n_{ver})

To connect the scalability measurement to end-to-end performance, we evaluate the hybrid latency model at a fixed verifier size ($n_{ver} = 16$), fixed bandwidth ($B = 5$ Mbps), and $p_{desync} = 0.1$, while varying the payload size from telemetry-scale to LiDAR-scale. Figure 5.4 reports the resulting expected latency.

As the compression ratio increases (fixed n_{ver} while n_{data} grows), the ordering between the two ID variants becomes consistent with the t_{ver} scalability results in Figure 5.3: the SHA-256-based scheme achieves lower end-to-end latency than RS-ID. Intuitively, at fixed n_{ver} and fixed p_{desync} , both schemes share essentially the same expected communication component (the verifier transmission on non-fallback ticks, and the same fallback transmission model); therefore, the gap between the curves in Figure 5.4 is dominated by the compute floor t_{ver} . Since SHA-256 scales with a lower per-byte cost than the current RS-ID implementation, it remains faster as n_{data} increases.

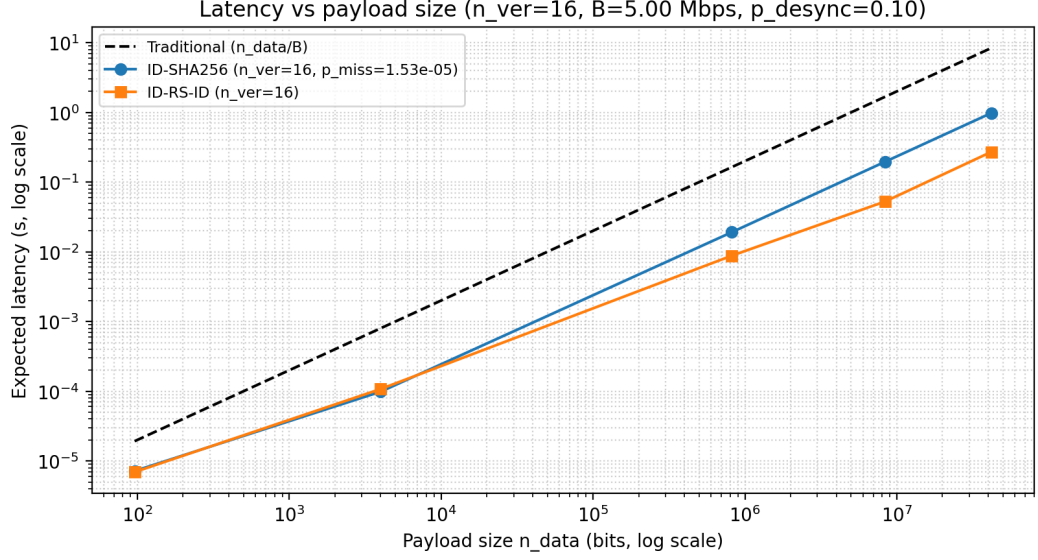


Figure 5.4: Expected latency vs. payload size ($n_{ver} = 16$, $B = 5$ Mbps, $p_{desync} = 0.1$). The ID-based schemes incur a compute floor (t_{ver}) but can reduce communication cost to n_{ver}/B on non-fallback ticks.

5.3 K-ID: Quantifying the Safety-Performance Trade-off

Having established the baseline performance, we now evaluate the **K-ID** extension, which introduces semantic tolerance (R) to filter benign deviations. We sweep the physical tolerance radius $R \in \{2, 5, 10, 20, 30\}$ and the verifier length $n_{ver} \in \{4, 8, 12, 16\}$ bits to map the design space.

5.3.1 Mechanism Analysis: The Tolerance-Verification Interaction

Figure 5.5 maps the full design space to decouple the roles of physical and cryptographic parameters. The heatmaps confirm a fundamental interplay: the physical radius R establishes the performance ceiling, as indicated by the speedup metrics, while the verifier length n_{ver} determines the residual safety floor.

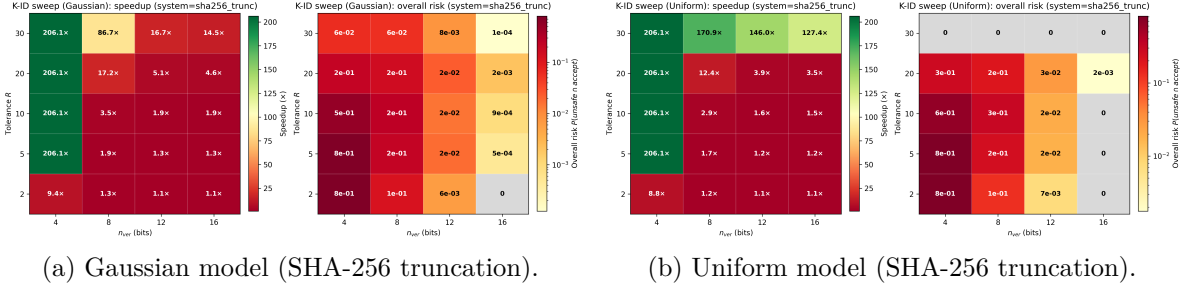


Figure 5.5: Heatmap summary of the K-ID grid over (R, n_{ver}) . The left panel in each figure shows speedup (linear scale); the right panel shows empirical overall risk $P(\text{unsafe} \cap \text{accept})$ (log-scaled). Cells marked with "0" indicate an empirical zero (no failures observed in simulation).

A critical insight from the Uniform model (Figure 5.5b) is the fundamental distinction between the two mechanisms that yield zero empirical risk. The top row of the heatmap ($R = 30$) demonstrates safety through physical immunity. Since this tolerance encapsulates the theoretical noise boundary ($a \approx 27.71$), the system relies on passive physical margins. Verification events are suppressed simply because the robot rarely violates the tolerance, resulting in massive speedups exceeding $127\times$ and zero risk due to protocol silence.

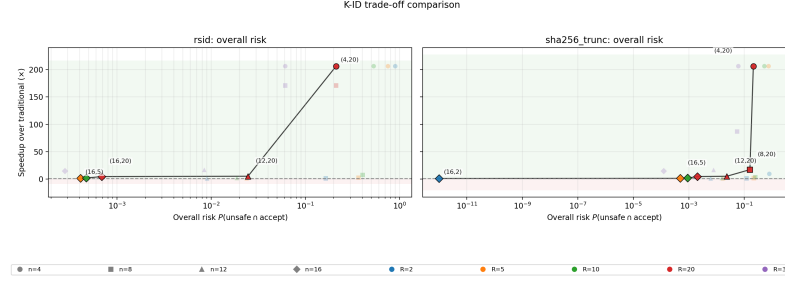
Conversely, the rightmost column ($n_{ver} = 16$ for radii $R < 20$) achieves safety through probabilistic immunity. In these configurations, the robot frequently exceeds the tighter tolerances, triggering continuous verification checks as indicated by the significantly lower speedups. However, despite the high frequency of checks, the 16-bit hash provides sufficient entropy to suppress collision probabilities below the simulation detection threshold, yielding an empirical risk of zero.

Between these two extremes lies a transitional “danger zone” around $R = 20$ when paired with weak verifiers. In this region, the tolerance is large enough to dilute the acceptance set $\mathcal{A}_K$ —thereby increasing the collision probability per check—but too small to contain the noise boundary. Consequently, the system suffers from the worst of both worlds: frequent verification triggers and a high likelihood of collision ($> 10^{-3}$), unless compensated by stronger verifiers ($n_{ver} \geq 12$).

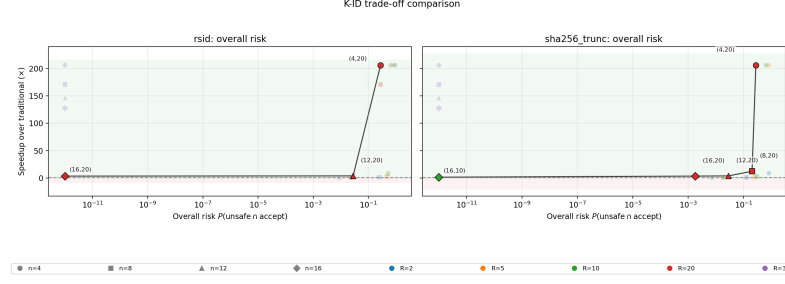
The Gaussian model (Figure 5.5a) presents a stricter challenge due to the distribution’s unbounded tails. Even at the maximum tolerance $R = 30$, the “Safe Harbor” is not absolute; residual risks in the range of 10^{-4} persist. This confirms that for unbounded noise processes, reliance on physical margins alone is insufficient. Instead, safety must be actively enforced by increasing the verifier length. The data demonstrates that increasing n_{ver} from 4 to 16 bits effectively suppresses the overall risk by orders of magnitude across all tolerance levels, with only a minor penalty in speedup due to the increased packet overhead.

5.3.2 Pareto Optimization for Deployment

To synthesize these multi-dimensional trade-offs into actionable design guidelines, we project the configuration grid onto a Speedup vs. Risk plane. Figure 5.6 visualizes the Pareto Frontier, connecting the non-dominated configurations that offer the maximum theoretical speedup for a given risk budget.



(a) Gaussian error model: Pareto comparison.



(b) Uniform error model: Pareto comparison.

Figure 5.6: Pareto frontier of **K-ID** design over (n_{ver}, R) . The solid black lines connect Pareto-nondominated configurations for the SHA-256 implementation, delineating the theoretical efficiency limit of the protocol. Points with $R = 30$ are shown for reference but are excluded from the Pareto-optimal set selection.

The Pareto frontiers in Figure 5.6 reveal that the optimal design strategy depends fundamentally on the noise containment regime.

For the SHA-256 implementation (Right Panels), the "Active Verification" regime ($R < 28$) forms a continuous curve in the lower-left region. Notably, the high-safety zone (Risk $< 10^{-3}$) is not populated solely by tight tolerances; rather, it includes configurations across the full range of active radii ($R \in \{2, 5, 10, 20\}$), provided they are paired with strong verifiers ($n_{ver} \geq 12$). In fact, the $R = 20$ configurations (red markers) frequently dominate the tighter tolerances in this region because their wider physical margins reduce transmission frequency while the strong hash maintains safety. The curve extends rightward as the verifier strength decreases, illustrating a smooth exchange of safety for marginal speedup gains.

A distinct "Passive Safety" regime appears at the far right, formed exclusively by the $R = 30$ configuration (purple marker). Because this tolerance exceeds the theoretical noise boundary, the protocol achieves a discontinuous jump in performance, with speedups exceeding $200\times$. However, unlike the active regime, this operating point incurs a fixed empirical risk floor determined by the noise distribution tails, which cannot be mitigated by increasing the verifier length.

Comparing the two algorithms, the SHA-256 frontier exhibits a significantly denser distribution of optimal points compared to RSID. In the active verification region, SHA-256 allows engineers to finely tune the verifier length to traverse the risk-speedup curve. In contrast, the RSID frontier (Left Panels) is sparse, often forcing a binary choice between ultra-conservative settings and the high-risk passive regime, thereby validating the architectural choice of truncated hashing for granular control.

5.4 Dynamic Evaluation Results: Random Walk with Drift

Building on the experimental design detailed in Section 4.5, this section presents the evaluation of the K-ID protocol under dynamic conditions. We first analyze the qualitative behavior of the protocol through a representative stress-test trace, followed by a quantitative statistical analysis of safety risks and bandwidth efficiency across the parameter space.

5.4.1 Qualitative Analysis: The Anatomy of Failure

To visualize the system's behavior under the stress-test configuration ($R = 10, n_{ver} = 12, \sigma = 1.0$), Figure 5.7 plots a 1,200-tick snapshot of the state evolution. The blue curve represents the robot's true state S_t undergoing a random walk, while the dashed orange line indicates the Digital Twin's estimate $\hat{S}_t$. The green shaded band denotes the semantic safety set $\mathcal{A}_K$.

The trace explicitly confirms the protocol's bandwidth suppression mechanism. For extended periods (e.g., $t \in [200, 500]$), the robot drifts within the safety envelope without triggering transmissions, effectively masking minor fluctuations. The green markers indicate nominal operation, where a boundary breach triggers an immediate fallback and correction. However, the red crosses highlight the critical failure mode induced by the 12-bit verifier: simultaneous boundary breaches and hash collisions. In these instances, the protocol fails to detect the excursion, allowing the robot to "escape" the safety set. This visualization validates the theoretical concern that probabilistic verification, when challenged by continuous drift, can lead to transient periods of unobserved instability.

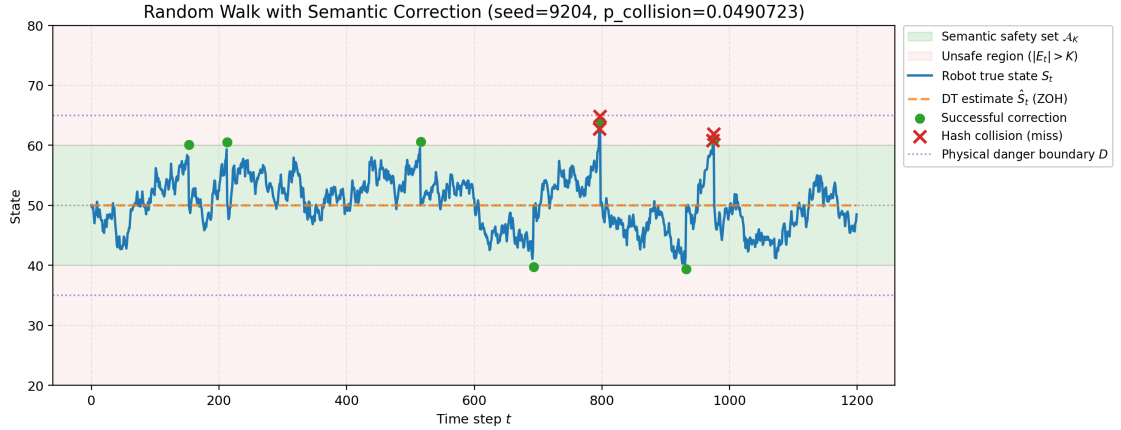


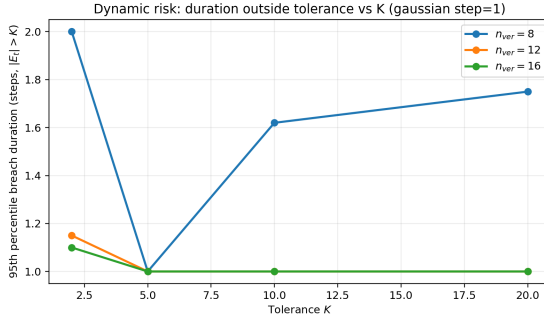
Figure 5.7: Drift-and-escape visualization for the dynamic random walk model ($R = 10, \sigma = 1.0$). The robot state S_t (blue) performs a random walk while the DT estimate $\hat{S}_t$ (orange) holds constant. Green dots mark successful corrections; red crosses mark collision-induced misses where the boundary breach is masked, allowing the trajectory to diverge beyond the safety set.

5.4.2 Quantitative Risk: Persistence and Magnitude

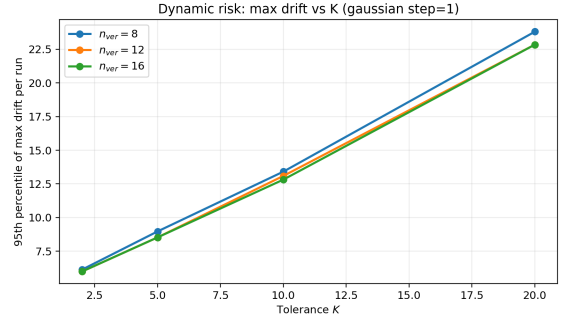
Beyond the single trace, we quantify the statistical risk profile using the parameter sweep data ($K \in \{2, 5, 10, 20\}$). Figure 5.8 reports the 95th-percentile summaries for Breach Duration (L_K) and Maximum Drift.

Figure 5.8(a) reveals the non-linear nature of dynamic risk. For weak verifiers ($n_{ver} = 8$, blue line), the breach duration exhibits a sharp jump as the tolerance K increases. This indicates that failures are not independent events; a hash collision at step t increases the likelihood of a collision at step $t + 1$ due to the high correlation of the drifting state. Consequently, a single miss often cascades into a persistent blind spot. In contrast, the $n_{ver} \geq 12$ configurations (orange and green lines) effectively suppress these consecutive failures, maintaining the breach duration near 1.0. This validates that while $n_{ver} = 12$ carries a marginal risk of single-step misses (as seen in the trace), it successfully prevents the catastrophic long-duration failures observed with 8-bit verifiers.

Figure 5.8(b) illustrates the physical magnitude of these failures. We observe a linear relationship between the tolerance K and the maximum peak drift. This confirms that while increasing K improves bandwidth efficiency (masking), it raises the stakes: when a failure eventually occurs—whether due to a miss or transmission latency—the accumulated error is proportional to the masking radius. Thus, the choice of K represents a direct trade-off between communication frequency and the worst-case physical deviation.



(a) 95th percentile Breach Duration L_K .



(b) 95th percentile of Maximum Drift.

Figure 5.8: Dynamic risk metrics vs. tolerance K (Gaussian drift $\sigma = 1.0$). Panel (a) shows that weak verifiers ($n = 8$) suffer from cascading failures as K increases, leading to long breach durations. Panel (b) shows that the peak physical error scales linearly with the masking radius K .

5.4.3 Efficiency Analysis: Inter-Fallback Times

Finally, we evaluate the bandwidth implications of the protocol through the distribution of **IFTs**. Figure 5.9 plots the empirical **CDF** of the silent intervals.

The results demonstrate a significant "right-shift" effect as the tolerance K increases. For the tightest tolerance ($K = 2$, blue line), the system transmits frequently, with approximately 99% of intervals being shorter than 10 ticks, indicating near-continuous communication. In contrast, for a relaxed tolerance ($K = 20$, red line), the curve flattens significantly, showing a median silence duration of approximately 50 ticks, with a "heavy tail" extending beyond 200 ticks. This statistical characteristic effectively transforms the continuous sensor stream into sporadic, bursty traffic. For network schedulers, the red curve implies that if $K = 20$ is configured, the channel remains idle for substantial periods, allowing the vast majority of time slots to be reallocated to other devices without contention.

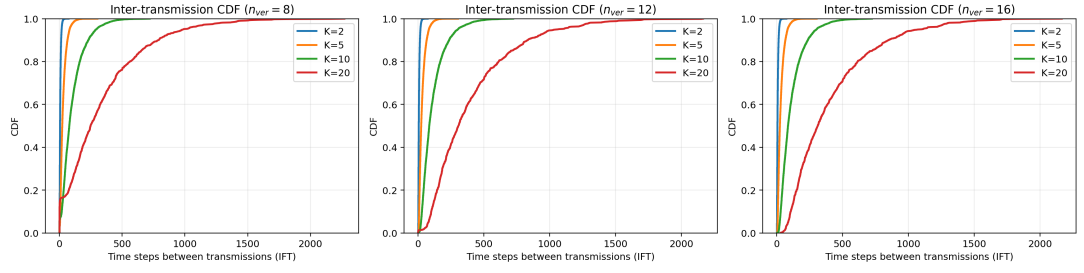


Figure 5.9: Empirical CDF of IFTs for the random-walk drift model. Larger K values shift the distribution to the right, indicating longer periods of silence (masking) and reduced channel utilization.

6 Conclusion and Future Work

6.1 Conclusion

This thesis establishes that **ID**-based synchronization is not merely a bandwidth-saving technique, but a latency-critical enabler for edge-based Digital Twins, particularly in bandwidth-constrained environments. By shifting the communication paradigm from "data reconstruction" to "state verification," we derive the following core conclusions:

First, the **ID** protocol effectively decouples synchronization latency from the state dimension. Experiments demonstrate that in bandwidth-limited regimes ($B < 1$ Mbps), **ID** offers a decisive latency advantage over traditional full-state transmission. Especially for high-dimensional sensor data such as **LiDAR**, although **ID** incurs a computational overhead, it significantly improves end-to-end freshness by avoiding unnecessary payload transmission.

Second, truncated cryptographic hashing via **SHA-256** represents the superior engineering choice over algebraic coding via **RS-ID** for edge deployments. While **RS-ID** offers unconditional information-theoretic security, our results reveal it suffers from severe field saturation effects at short blocklengths ($n_{ver} = 4$) and poor scalability with payload size. In contrast, **SHA-256** maintains a stable base encoding latency of approximately $2.2 \mu s$ with robust scalability. Consequently, unless information-theoretic secrecy is explicitly required, **SHA-256** better balances computational speed with collision robustness for **URLLC** applications.

Third, **K-ID** successfully realizes the paradigm shift from "bitwise exactness" to "semantic compliance." The Pareto frontier analysis proves that by introducing semantic tolerance (K), the system can leverage the physical stability margins of the control loop to trade for communication efficiency. Experimental data shows that a moderate tolerance (e.g., $R = 10$ within a 100-unit range) reduces the fallback frequency by an order of magnitude while maintaining a bounded conditional miss probability. This validates **K-ID**'s ability to filter benign jitter and consume bandwidth only for significant state deviations.

Fourth, dynamic evaluation reveals the time-correlated risk of "cascading failures." Unlike static models, the random walk experiments show that weak verifiers ($n_{ver} = 8$) can lead to single hash collisions evolving into persistent, undetected boundary breaches. Furthermore, the heavy-tail distribution of Inter-Fallback Times (IFT) confirms that increasing K effectively transforms continuous sensor streams into sporadic, bursty traffic, freeing up substantial channel resources. However, this bandwidth dividend trades off against physical risk, as the peak physical drift during a failure event scales linearly with the masking radius K .

In summary, by establishing a quantifiable trade-off model among computation, communication, and control fidelity, the **K-ID** protocol provides a theoretical foundation and engineering guidelines for the design of next-generation semantic communication networks.

6.2 Future Work

While this work has characterized the latency and reliability of **K-ID**, several avenues remain to extend its theoretical depth and practical applicability in next-generation networks.

6.2.1 Energy-Aware Identification Policies

Current evaluations utilize traffic volume and latency as proxies for system efficiency. However, for battery-powered **Internet of Things (IoT)** devices, energy consumption is the ultimate constraint. Future work should explicitly model the energy–delay product, balancing the computational energy cost of hashing against the transmission energy cost of radio interfaces. By integrating the First-Order Radio Model [9] with real-time power profiling of edge hardware, we aim to derive an energy-optimal policy that dynamically adapts the tolerance K and verifier length n_{ver} based on the device’s remaining battery life—tightening safety margins when energy is abundant and relaxing them to maximize operational duration under critical battery levels.

6.2.2 Integration with 6G Physical Layer Security

The current implementation relies on synchronized **PRNGs**, which introduces a dependency on upper-layer clock synchronization protocols like **PTP**. To align with the architecture of 6G networks, future iterations will investigate the integration of **PLKG**. By exploiting the channel reciprocity in **Time-Division Duplex (TDD)** systems, the robot and the **DT** can extract common randomness directly from the **Channel State Information (CSI)**. This approach would turn channel noise into a synchronization resource, eliminating the need for clock synchronization and enhancing robustness against timing attacks in harsh industrial environments.

6.2.3 Scalability to Multi-Agent Systems

This thesis focused on a single robot–**DT** loop. However, industrial scenarios often involve swarms of robots sharing a physical workspace. A critical extension is the development of semantic multiple access schemes, where multiple robots superimpose their identification codes on a shared channel. Future work will explore algebraic constructions that allow the **DT** to verify the aggregate safety of the swarm directly in the coded domain, checking for "Semantic Inter-Robot Collisions" without decoding individual states, thereby scaling the **K-ID** paradigm to massive multi-agent systems.

6.2.4 Predictive Digital Twins and Adaptive K-ID

The dynamic drift evaluation in this thesis uses a deliberately simple predictor (ZOH) to isolate protocol-induced effects. A natural next step is to integrate stronger predictors such as Kalman filtering or learned kinematic models, and to study how prediction quality reshapes both the masking risk (through the breach rate) and the collision risk (through the verification rate). In addition, future work should explore adaptive policies that tune (K, n_{ver}) online based on observed breach statistics and safety budgets, tightening the tolerance under persistent drift and relaxing it during stable operation to reduce bandwidth.

6.2.5 Decoder Compute, Scheduling, and Quantization Sensitivity

Beyond bandwidth and latency, practical deployments must also account for DT-side compute and memory. Since the acceptance set size scales as $|\mathcal{A}_K| \approx 2\lfloor K/\Delta \rfloor + 1$, a direct implementation precomputes $\mathcal{V}_K$ by hashing $\mathcal{O}(|\mathcal{A}_K|)$ states per DT update, and stores up to $|\mathcal{V}_K| \leq |\mathcal{A}_K|$ verifier codewords. Future work should therefore characterize end-to-end performance under explicit decoder compute constraints (e.g., bounded hash evaluations per tick), and study algorithmic accelerations (incremental updates, caching, GPU offload).

In addition, the dynamic setting naturally suggests a scheduling-centric view: rather than only reporting per-tick risk, one can characterize the distribution of inter-transmission times (survival/CDF metrics) to answer questions like “with 99% probability, how long does the channel remain idle?” Finally, because Δ is tied to sensor precision and DT computational budget, a systematic sensitivity analysis over Δ is needed to guide robust configuration across heterogeneous sensing modalities.

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A Appendix

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